

Discussion: Identifiability and Optimization in Diagonal Kalman Delta Networks

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1 Setup

The diagonal Kalman Delta Network (KDN) predicts a diagonal covariance and uses it to form an uncertainty-dependent delta-rule write. For key $\mathbf{k}_t$, transition $\boldsymbol{\alpha}_t$, process noise $\boldsymbol{\omega}_t$, and observation noise r_t ,

$$\hat{\mathbf{p}}_t = \boldsymbol{\alpha}_t^2 \odot \mathbf{p}_{t-1} + \boldsymbol{\omega}_t, \quad (1)$$

$$\boldsymbol{\kappa}_t = \frac{\hat{\mathbf{p}}_t \odot \mathbf{k}_t}{r_t + \sum_i \hat{p}_{t,i} k_{t,i}^2}. \quad (2)$$

The memory update remains a delta update,

$$\mathbf{S}_t = \mathbf{D}_t \mathbf{S}_{t-1} + \boldsymbol{\kappa}_t \left(\mathbf{v}_t - (\mathbf{D}_t \mathbf{S}_{t-1})^\top \mathbf{k}_t \right)^\top, \quad \mathbf{D}_t = \text{diag}(\boldsymbol{\alpha}_t). \quad (3)$$

The implementation additionally tracks diagonal information $\mathbf{c}_t = 1/\mathbf{p}_t$ with the calibrated projected update

$$\mathbf{c}_t = \frac{1}{\hat{\mathbf{p}}_t} + \frac{d_k}{r_t} \mathbf{k}_t^2. \quad (4)$$

This is a scan-compatible diagonal-information approximation, not an exact diagonal Kalman posterior.

2 The Scale Symmetry

For any constant $\lambda > 0$, consider

$$\mathbf{p}'_t = \lambda \mathbf{p}_t, \quad \boldsymbol{\omega}'_t = \lambda \boldsymbol{\omega}_t, \quad r'_t = \lambda r_t, \quad \mu' = \frac{\mu}{\lambda}, \quad (5)$$

where $\mu = \mathbf{c}_0$ is the initial information and hence $\mathbf{p}_0 = 1/\mu$. The predicted covariance scales as

$$\hat{\mathbf{p}}'_t = \boldsymbol{\alpha}_t^2 \odot (\lambda \mathbf{p}_{t-1}) + \lambda \boldsymbol{\omega}_t = \lambda \hat{\mathbf{p}}_t. \quad (6)$$

Consequently, the gain is invariant:

$$\boldsymbol{\kappa}'_t = \frac{\lambda \hat{\mathbf{p}}_t \odot \mathbf{k}_t}{\lambda r_t + \sum_i \lambda \hat{p}_{t,i} k_{t,i}^2} = \boldsymbol{\kappa}_t. \quad (7)$$

The information update also transforms consistently as $\mathbf{c}'_t = \mathbf{c}_t/\lambda$. Therefore $\boldsymbol{\omega}$, r , and μ contain a common covariance-scale degree of freedom that cannot be identified from the memory output. Positive floors weaken the symmetry near their boundaries but do not remove the general conditioning problem.

The transition $\boldsymbol{\alpha}_t$ is different. It affects both covariance prediction and the memory transition $\mathbf{D}_t \mathbf{S}_{t-1}$, so it is not part of the exact scale symmetry. Nevertheless, $\boldsymbol{\alpha}_t$ and $\boldsymbol{\omega}_t$ are partially confounded in the covariance path because multiple pairs can produce similar $\hat{\mathbf{p}}_t$.

3 What the Noisy-MQAR Results Establish

The original diagonal model obtains 76.2 ± 22.3 percent over the 5×5 initialization and data grid. Its initialization component of variation (19.2 points) is larger than its data component (11.4 points). This is not uniform under-capacity: many cells reach 88–96 percent, while a few fail severely.

The most informative intervention is isotropic gain initialization. Zeroing the output map of the process-noise projection makes $\boldsymbol{\omega}_t$ uniform across channels at initialization, after which anisotropy is learned. At the same learning rate and with the same optimizer, this changes performance from

$$76.2 \pm 22.3 \quad \longrightarrow \quad 83.7 \pm 10.4, \quad (8)$$

and halves the initialization variation from 19.2 to 9.0 points. It also rescues both severe collapse cells. Thus the current evidence identifies random initial gain anisotropy as the primary failure, not the nominal learning rate or scale symmetry.

The scale symmetry remains relevant as a secondary optimization defect. It creates a flat direction, gives equivalent physical models different raw parameter values, and lets coupled quantities evolve under different optimizer geometries. In the current Muon setup, matrix weights use Muon with adjusted learning rates, while biases and the one-dimensional prior parameter use AdamW. A single printed learning rate therefore does not imply uniform motion of the physical uncertainty variables.

The prior μ is unlikely to explain late failures. It only controls the initial information, and its influence should decay rapidly over a long sequence when process noise has a positive floor. Learning it adds a weakly identified parameter with little expected benefit.

4 Ways to Remove or Reduce the Symmetry

4.1 Fix the observation-noise scale

The strongest and simplest gauge choice is

$$r_t \equiv 1, \quad \mu \equiv 1. \quad (9)$$

The model then learns process uncertainty in units of observation noise:

$$\boldsymbol{\omega}_t = \boldsymbol{\omega}_{\min} + \text{softplus}(f_{\omega}(\mathbf{x}_t)). \quad (10)$$

Because r can no longer be rescaled, the common covariance-scale symmetry is removed. This parameterization is the preferred baseline unless token-dependent observation reliability is shown empirically to matter.

4.2 Learn bounded relative observation reliability

If heteroscedastic observation noise is useful, it should change confidence relative to a fixed reference rather than learn an unrestricted global scale. One option is

$$r_t = \exp(s \tanh f_r(\mathbf{x}_t)), \quad (11)$$

with fixed s . For $s = \log 4$, $r_t \in [1/4, 4]$. The projection should have no learnable global bias, and the prior can remain fixed at $\mu = 1$. A soft anchoring penalty can further control drift,

$$\mathcal{L}_{\text{scale}} = \gamma (\mathbb{E}[\log r_t])^2. \quad (12)$$

This reduces rather than mathematically eliminates every possible input-dependent rescaling, but it prevents the practically harmful unbounded common-scale direction.

4.3 Parameterize a stationary uncertainty level

The α - ω interaction can be made more interpretable with

$$\omega_{t,i} = \omega_{\min} + (1 - \alpha_{t,i}^2) \rho_{t,i}, \quad \rho_{t,i} = \text{softplus}(f_\rho(\mathbf{x}_t)). \quad (13)$$

Here α controls retention and ρ is the uncertainty level toward which prediction moves. This is exactly the completed **DiagKLA1** experiment. Coupling alone changes 76.2 ± 22.3 to 76.9 ± 20.9 , which is effectively a wash. The completed **DiagKLA1-zi** experiment combines coupling with isotropic gain initialization and obtains 80.1 ± 13.3 , below the simpler **DiagKLA-zi** result of 83.7 ± 10.4 . Thus this parameterization has already been tested: it neither replaces isotropic initialization nor improves on free additive process noise once that initialization is used.

5 Recommended Ablation

The transition-coupled process-noise comparison is already complete:

Variant	Free additive ω	$(1 - \alpha^2)\rho$
Random anisotropic initialization	76.2 ± 22.3	76.9 ± 20.9
Isotropic gain initialization	83.7 ± 10.4	80.1 ± 13.3

The remaining experiment should isolate covariance-scale choices while keeping free additive ω and isotropic gain initialization fixed:

Variant	μ	r_t	ω_t
A	learned	learned positive	free additive
B	fixed at 1	learned positive	free additive
C	fixed at 1	fixed at 1	free additive
D	fixed at 1	bounded relative	free additive

Run at least the same 5×5 initialization–data grid. Report mean, total standard deviation, initialization variation, non-finite runs, and convergence time. During training, $\log \omega$, r , effective write strength, gain anisotropy, the minimum channel gain, and gradient norms for the process-noise, observation-noise, prior, and transition parameters.

If C matches or improves A, the production model should remove both r_{proj} and μ_{param} . If D improves C, retain only bounded relative observation reliability. Only after fixing this gauge is a separate uncertainty-parameter learning-rate sweep readily interpretable.

6 Why DiagKLA Does Not Yet Dominate IsoKLA

The available implementation tests establish that the Triton kernels reproduce the intended PyTorch recurrence; they do not establish that this recurrence is the best approximation for an associative memory. Empirically, the extra diagonal uncertainty state has not produced a general advantage over the isotropic state. This section separates the approximation question from the kernel-correctness question and states the experiment needed to identify the useful part of the diagonal model.

6.1 Current evidence

The controlled 220M/20B information-scale runs do not establish a general DiagKLA advantage. The best default-calibrated variants give:

Metric	DiagKLA, $s = d_k$	IsoKLA, $s = 1$
WikiText word perplexity ↓	28.99	28.79
LAMBADA perplexity ↓	35.12	35.66
Five-task multiple-choice average ↑	0.472	0.477
S-NIAH-2 at 4K ↑	0.272	0.046
S-NIAH-3 at 2K ↑	0.004	0.008
MK-NIAH-1 at 4K ↑	0.178	0.182

General language-modeling quality is mixed and effectively tied. DiagKLA has a large, specific S-NIAH-2 advantage, a direction that also appears in the earlier 220M comparison, but it does not improve the harder single-key or multikey probes. The recent sweeps use one training seed and different GPU world sizes, and a current-code 750M DiagKLA evaluation is not yet available. The evidence therefore supports a task-specific single-key retrieval signal, not a general DiagKLA ranking above IsoKLA.

The noisy-MQAR grid gives a similar result. Uniform process-noise initialization raises DiagKLA from 76.2 ± 22.3 to 83.7 ± 10.4 , while the best stabilized IsoKLA variant obtains 83.0 ± 8.4 . These are effectively tied at this resolution. The older trained-gain probe that reports large off-key gain and over-writing predates the current initialization and memory kernel; it motivates the mechanism test below but should not be treated as a diagnosis of the current default.

6.2 Approximation and calibration

With predicted diagonal covariance $\hat{\mathbf{P}}_t = \text{diag}(\hat{\mathbf{p}}_t)$, the exact Gaussian measurement update produces a dense posterior whose diagonal is

$$p_{t,i}^{\text{exact}} = \hat{p}_{t,i} - \frac{\hat{p}_{t,i}^2 k_{t,i}^2}{r_t + \sum_j \hat{p}_{t,j} k_{t,j}^2}. \quad (14)$$

The scan-compatible DiagKLA state instead uses

$$p_{t,i}^{\text{proj}} = \left(\hat{p}_{t,i}^{-1} + s \frac{k_{t,i}^2}{r_t} \right)^{-1}, \quad s = d_k \text{ by default.} \quad (15)$$

Even at $s = 1$, Equation (15) is not the diagonal of the exact posterior in Equation (14). For $s \neq 1$, the gain and information update additionally correspond to different measurement scales: the

current gain retains precision $1/r_t$, while the state used by future gains receives precision s/r_t . Thus $s = d_k$ is a dimension calibration of a gain-conditioning state, not an exact Kalman consequence.

For a spread-out unit key, $\mathbb{E}[k_i^2] = 1/d_k$, so $s = d_k$ gives a unit expected per-coordinate increment. Averaging these calibrated coordinates recovers the IsoKLA scale,

$$\frac{1}{d_k} \sum_i d_k \frac{k_{t,i}^2}{r_t} = \frac{\|\mathbf{k}_t\|_2^2}{r_t} = \frac{1}{r_t}. \quad (16)$$

Matching mean information does not match covariance or gain because inversion is nonlinear. Moreover, for a uniformly distributed unit key,

$$\text{Var}(d_k k_i^2) = \frac{2(d_k - 1)}{d_k + 2} \longrightarrow 2. \quad (17)$$

The coordinate noise therefore does not vanish with width. DiagKLA can interpret coordinate energy fluctuations as persistent uncertainty structure. The projection is also basis dependent: $(1, 1)/\sqrt{2}$ and $(1, -1)/\sqrt{2}$ are orthogonal but have identical elementwise squares and hence identical information increments. Whether the learned key basis aligns this diagonal statistic with useful uncertainty is an empirical question.

6.3 From exact Gaussian filtering to an online variational Möbius state

At $s = 1$, the projected rule is exactly a mean-field variational-inference update, not merely a Gaussian-inspired heuristic. Each token defines a dense Gaussian filtering posterior, and that posterior is projected onto a diagonal Gaussian family by minimizing the reverse KL divergence. Repeating this projection after every token gives an online variational filter. The production factor $s = d_k$ is a separate calibration applied after this variational derivation; it is not the optimum of the same variational objective.

To make the variational step explicit, use the memory notation of the main paper. Let $\mathbf{S}_t^* \in \mathbb{R}^{d_k \times d_v}$ be the random latent associative memory and let $\mathbf{S}_t$ be its posterior-mean estimate. Fix a value coordinate j , and denote column j of these matrices by $\mathbf{s}_{t,j}^*$ and $\mathbf{s}_{t,j}$, respectively. Thus the latent vector in the following scalar observation model is not a new state: it is one column of $\mathbf{S}_t^*$. Its predicted model is

$$\mathbf{s}_{t,j}^* | \mathcal{F}_{t-1} \sim \mathcal{N}(\widehat{\mathbf{s}}_{t,j}, \widehat{\mathbf{P}}_t), \quad v_{t,j} = \mathbf{k}_t^\top \mathbf{s}_{t,j}^* + e_{t,j}, \quad e_{t,j} \sim \mathcal{N}(0, r_t), \quad (18)$$

where $\widehat{\mathbf{s}}_{t,j}$ is column j of the predicted mean $\widehat{\mathbf{S}}_t$. Let $\pi_{t,j}$ denote the exact one-step posterior density of $\mathbf{s}_{t,j}^*$ after observing $v_{t,j}$. Its negative log density, evaluated at a possible column value $\mathbf{s}$, has quadratic part

$$-\log \pi_{t,j}(\mathbf{s}) = \frac{1}{2}(\mathbf{s} - \widehat{\mathbf{s}}_{t,j})^\top \widehat{\mathbf{P}}_t^{-1}(\mathbf{s} - \widehat{\mathbf{s}}_{t,j}) + \frac{(v_{t,j} - \mathbf{k}_t^\top \mathbf{s})^2}{2r_t} + \text{const}. \quad (19)$$

Its Hessian gives the exact posterior information matrix

$$\mathbf{C}_t^* = (\mathbf{P}_t^*)^{-1} = \widehat{\mathbf{P}}_t^{-1} + \frac{1}{r_t} \mathbf{k}_t \mathbf{k}_t^\top. \quad (20)$$

Applying the Woodbury identity to Equation (20) recovers Equation (14). Its mean is

$$\mathbf{s}_{t,j} = \widehat{\mathbf{s}}_{t,j} + \boldsymbol{\kappa}_t \left(v_{t,j} - \mathbf{k}_t^\top \widehat{\mathbf{s}}_{t,j} \right), \quad (21)$$

which is column j of the matrix update for $\mathbf{S}_t$ in the setup section. Thus $\mathbf{S}_t$ is the point estimate (posterior mean), whereas $\mathbf{P}_t^\star$ describes the uncertainty around each of its columns. Although this one-step posterior is analytically tractable, carrying its dense covariance would destroy the independent per-coordinate state used by the associative scan. Variational inference enters here as a structured computational approximation.

Choose the mean-field Gaussian family for this memory column,

$$\mathcal{Q} = \{q_{t,j}(\mathbf{s}) = \mathcal{N}(\boldsymbol{\mu}_{t,j}, \text{diag}(\mathbf{p}_t)) : \mathbf{p}_t > 0\}. \quad (22)$$

For the one-token model in Equation (18), its ELBO is

$$\mathcal{L}_{t,j}(q_{t,j}) = \mathbb{E}_{q_{t,j}}[\log p(v_{t,j} \mid \mathbf{s}_{t,j}^\star)] - \text{KL}(q_{t,j}(\mathbf{s}_{t,j}^\star) \parallel p(\mathbf{s}_{t,j}^\star \mid \mathcal{F}_{t-1})). \quad (23)$$

The standard variational identity gives

$$\log p(v_{t,j} \mid \mathcal{F}_{t-1}) = \mathcal{L}_{t,j}(q_{t,j}) + \text{KL}(q_{t,j} \parallel \pi_{t,j}). \quad (24)$$

Therefore the variational update can be written equivalently as

$$q_{t,j}^{\text{VI}} = \arg \max_{q_{t,j} \in \mathcal{Q}} \mathcal{L}_{t,j}(q_{t,j}) = \arg \min_{q_{t,j} \in \mathcal{Q}} \text{KL}(q_{t,j} \parallel \pi_{t,j}). \quad (25)$$

Because the target $\pi_{t,j} = \mathcal{N}(\mathbf{s}_{t,j}, \mathbf{P}_t^\star)$ is Gaussian, this KL reduces, up to terms independent of $\boldsymbol{\mu}_{t,j}$ and $\mathbf{p}_t$, to

$$\text{KL}(q_{t,j} \parallel \pi_{t,j}) = \frac{1}{2}(\boldsymbol{\mu}_{t,j} - \mathbf{s}_{t,j})^\top \mathbf{C}_t^\star (\boldsymbol{\mu}_{t,j} - \mathbf{s}_{t,j}) + \frac{1}{2} \sum_i (C_{t,ii}^\star p_{t,i} - \log p_{t,i}) + \text{const}. \quad (26)$$

Since $\mathbf{C}_t^\star$ is positive definite, the mean optimum is the exact filtering mean, $\boldsymbol{\mu}_{t,j} = \mathbf{s}_{t,j}$. The variance objective separates over key coordinates i , and

$$\frac{\partial}{\partial p_{t,i}} \text{KL}(q_{t,j} \parallel \pi_{t,j}) = \frac{1}{2} \left(C_{t,ii}^\star - \frac{1}{p_{t,i}} \right), \quad \frac{\partial^2}{\partial p_{t,i}^2} \text{KL}(q_{t,j} \parallel \pi_{t,j}) = \frac{1}{2p_{t,i}^2} > 0. \quad (27)$$

The positive second derivative shows that the stationary point is the unique minimum. Setting the first derivative to zero gives $C_{t,ii}^\star = 1/p_{t,i}$ and hence

$$p_{t,i}^{\text{rev}} = \frac{1}{C_{t,ii}^\star} = \left(\widehat{p}_{t,i}^{-1} + \frac{k_{t,i}^2}{r_t} \right)^{-1}. \quad (28)$$

The final equality follows directly from the diagonal of the exact information update, $C_{t,ii}^\star = \widehat{p}_{t,i}^{-1} + k_{t,i}^2/r_t$. Thus Equation (15) at $s = 1$ is exactly the reverse-KL mean-field variational optimum: diagonalize the posterior precision and then invert. Using the previous diagonal approximation to form the next predictive prior and repeating this ELBO optimization token by token gives online mean-field variational inference. It is a reverse-KL project-after-each-token filter, rather than a single batch approximation to the full latent trajectory.

For comparison, the forward-KL, or moment, projection onto the same diagonal family satisfies

$$\arg \min_{\mathbf{p} > 0} \text{KL}(\pi_{t,j} \parallel q_{t,j}) \implies p_{t,i} = P_{t,ii}^\star = p_{t,i}^{\text{exact}}. \quad (29)$$

This inclusive-KL projection matches the exact marginal variances, but it is not the ELBO/reverse-KL variational objective above. The two projections differ because diagonalization and inversion do not commute,

$$\frac{1}{C_{t,ii}^*} \neq [(C_t^*)^{-1}]_{ii} = P_{t,ii}^* \quad \text{in general.} \quad (30)$$

For a Gaussian posterior these two quantities have different meanings:

$$\frac{1}{C_{t,ii}^*} = \text{Var}_{\pi_{t,j}}((s_{t,j}^*)_i \mid (s_{t,j}^*)_{-i}), \quad P_{t,ii}^* = \text{Var}_{\pi_{t,j}}((s_{t,j}^*)_i). \quad (31)$$

The reverse-KL diagonal approximation therefore carries conditional variances, as if the other coordinates were known, rather than their larger marginal variances. This is why it is generally overconfident when the exact posterior contains cross-coordinate correlations. Indeed, Equation (67) shows that marginalizing the other coordinates replaces r_t in Equation (28) by $r_t + \sum_{\ell \neq i} \hat{p}_{t,\ell} k_{t,\ell}^2$. Consequently $p_{t,i}^{\text{rev}} \leq p_{t,i}^{\text{exact}}$, with equality if $k_{t,i} = 0$ or if the other coordinates contribute no key-weighted uncertainty. The selected projection is therefore systematically no less confident than the exact marginal.

The computational payoff of this mean-field variational family appears after returning to information variables. Define

$$c_{t-1,i} = p_{t-1,i}^{-1}, \quad u_{t,i} = s \frac{k_{t,i}^2}{r_t}. \quad (32)$$

Diagonal prediction followed by the projected measurement update gives

$$\hat{c}_{t,i} = \frac{c_{t-1,i}}{\alpha_{t,i}^2 + \omega_{t,i} c_{t-1,i}}, \quad (33)$$

$$c_{t,i} = \hat{c}_{t,i} + u_{t,i} = \frac{(1 + u_{t,i} \omega_{t,i}) c_{t-1,i} + u_{t,i} \alpha_{t,i}^2}{\omega_{t,i} c_{t-1,i} + \alpha_{t,i}^2}. \quad (34)$$

Each token therefore supplies an input-only 2×2 matrix

$$\mathbf{M}_{t,i} = \begin{pmatrix} 1 + u_{t,i} \omega_{t,i} & u_{t,i} \alpha_{t,i}^2 \\ \omega_{t,i} & \alpha_{t,i}^2 \end{pmatrix}, \quad u_{t,i} = s \frac{k_{t,i}^2}{r_t}. \quad (35)$$

Möbius maps compose by matrix multiplication, so the prefix product $\mathbf{M}_{t,i} \cdots \mathbf{M}_{1,i}$ supplies every information state by an associative scan, independently for each channel. The production kernel implements this as chunk-local map composition, a short carry scan, and a chunk-local emit pass. The cross-channel sum in the current gain is evaluated only after the exclusive information prefixes are available, so it is a token-local reduction rather than part of the recurrence. By contrast, the moment projection makes each coordinate's information increment depend on all the other predicted variances, as in Equation (67); it retains $O(d_k)$ state but loses these independent input-only Möbius maps.

The variational optimum above has $s = 1$. Production deliberately departs from that optimum and uses $s = d_k$ as a width calibration. The keys are ℓ_2 normalized, so

$$\frac{1}{d_k} \sum_i k_{t,i}^2 = \frac{1}{d_k}, \quad \mathbb{E}[k_{t,i}^2] = \frac{1}{d_k} \quad \text{for exchangeable coordinates.} \quad (36)$$

The literal projection therefore adds only $1/(d_k r_t)$ information to a typical coordinate. Standardizing a coordinate to unit second moment, $\tilde{k}_{t,i} = \sqrt{d_k} k_{t,i}$, gives

$$u_{t,i} = \frac{\tilde{k}_{t,i}^2}{r_t} = d_k \frac{k_{t,i}^2}{r_t}, \quad (37)$$

which is precisely $s = d_k$. Equivalently, in the symmetric case $\hat{p}_{t,i} = \hat{p}_t$ and $k_{t,i}^2 = 1/d_k$,

$$p_{t,i}^{\text{proj},s} = \frac{\hat{p}_t}{1 + (s/d_k)\hat{p}_t/r_t}. \quad (38)$$

Matching the scalar normalized-key update $(\hat{p}_t^{-1} + r_t^{-1})^{-1}$ selects $s = d_k$. This is the same mean-scale matching stated in Equation (16), now expressed as a feature standardization and a covariance-timescale match.

This calibration is not variational inference for the original observation model. At $s = d_k$, the total added diagonal precision is d_k/r_t , whereas the trace of the exact rank-one increment is only $1/r_t$. One could obtain the scaled precision from the power-posterior ELBO for memory column j ,

$$\mathcal{L}_{t,j}^{(s)}(q_{t,j}) = s \mathbb{E}_{q_{t,j}}[\log p(v_{t,j} \mid \mathbf{s}_{t,j}^*)] - \text{KL}(q_{t,j}(\mathbf{s}_{t,j}^*) \parallel p(\mathbf{s}_{t,j}^* \mid \mathcal{F}_{t-1})), \quad (39)$$

which is equivalent to replacing r_t by r_t/s . Such a coherent tempered variational update would use r_t/s in both the current gain and the posterior covariance. DiagKLA instead keeps r_t in the current gain and applies s only to the information state passed forward.

The production formulation should therefore be read as two distinct operations: (i) at $s = 1$, a reverse-KL mean-field variational projection that preserves a positive $O(d_k)$ state and an associative gain scan; and (ii) after that derivation, a d_k coordinate calibration that makes the future-confidence timescale independent of head width. The resulting state is a calibrated gain-conditioning state, not the ordinary-ELBO covariance—or exact covariance—of one coherent linear-Gaussian model. Equation (55) below gives a generalized variational objective for this deliberate separation.

6.4 Information scale separates present writing from future protection

The role of s is clearest after separating the memory update at token t from the uncertainty state passed to later tokens. Define the key-weighted predicted variance and the *effective write strength*

$$z_t = \sum_i \hat{p}_{t,i} k_{t,i}^2, \quad \beta_t^{\text{eff}} = \mathbf{k}_t^\top \boldsymbol{\kappa}_t. \quad (40)$$

Contracting the diagonal gain with its own key evaluates β_t^{eff} in closed form,

$$\beta_t^{\text{eff}} = \sum_i k_{t,i} \frac{\hat{p}_{t,i} k_{t,i}}{r_t + \sum_j \hat{p}_{t,j} k_{t,j}^2} = \frac{\sum_i \hat{p}_{t,i} k_{t,i}^2}{r_t + z_t} = \frac{z_t}{r_t + z_t}, \quad 1 - \beta_t^{\text{eff}} = \frac{r_t}{r_t + z_t}. \quad (41)$$

Positive variances and observation noise give $z_t \geq 0$ and $r_t > 0$, hence $\beta_t^{\text{eff}} \in [0, 1)$.

Residual contraction. Write $\tilde{\mathbf{S}}_t = \mathbf{D}_t \mathbf{S}_{t-1}$ for the decayed memory and $\mathbf{e}_t = \mathbf{v}_t - \tilde{\mathbf{S}}_t^\top \mathbf{k}_t$ for the prediction error that the delta rule corrects, so that the memory recurrence is the rank-one update

$$\mathbf{S}_t = \tilde{\mathbf{S}}_t + \boldsymbol{\kappa}_t \mathbf{e}_t^\top. \quad (42)$$

Transposing Equation (42) gives $\mathbf{S}_t^\top = \tilde{\mathbf{S}}_t^\top + \mathbf{e}_t \boldsymbol{\kappa}_t^\top$. Reading the updated memory at the same key, the scalar $\boldsymbol{\kappa}_t^\top \mathbf{k}_t = \beta_t^{\text{eff}}$ factors out of the rank-one term:

$$\mathbf{S}_t^\top \mathbf{k}_t = \tilde{\mathbf{S}}_t^\top \mathbf{k}_t + \mathbf{e}_t (\boldsymbol{\kappa}_t^\top \mathbf{k}_t) = \tilde{\mathbf{S}}_t^\top \mathbf{k}_t + \beta_t^{\text{eff}} \mathbf{e}_t, \quad (43)$$

$$\mathbf{v}_t - \mathbf{S}_t^\top \mathbf{k}_t = \underbrace{(\mathbf{v}_t - \tilde{\mathbf{S}}_t^\top \mathbf{k}_t)}_{=\mathbf{e}_t} - \beta_t^{\text{eff}} \mathbf{e}_t = (1 - \beta_t^{\text{eff}}) \mathbf{e}_t = \frac{r_t}{r_t + z_t} \mathbf{e}_t. \quad (44)$$

Thus β_t^{eff} is the fraction of the current residual corrected at its own key. Equation (44) is the delta-rule instance of the standard Kalman innovation contraction $\mathbf{I} - \mathbf{K}_t \mathbf{H}_t$, which for the scalar-key observation $\mathbf{H}_t = \mathbf{k}_t^\top$ collapses to the scalar $1 - \mathbf{k}_t^\top \boldsymbol{\kappa}_t$. Vanilla DeltaNet is the special case $\boldsymbol{\kappa}_t = \beta_t \mathbf{k}_t$ with $\|\mathbf{k}_t\|_2 = 1$, which recovers $\beta_t^{\text{eff}} = \beta_t$; here the same quantity is instead an emergent function $z_t/(r_t + z_t)$ of the uncertainty state.

Two properties of Equation (44) are used below. First, the same $\boldsymbol{\kappa}_t$ controls both the value write $+\boldsymbol{\kappa}_t \mathbf{v}_t^\top$ and the erasure $-\boldsymbol{\kappa}_t \mathbf{k}_t^\top \tilde{\mathbf{S}}_t$, so a gain that assimilates the current observation more strongly necessarily also erases more of what the memory already held at that key. Second, the contraction depends on $\boldsymbol{\kappa}_t$ only through its component along $\mathbf{k}_t$: any off-key part $\boldsymbol{\kappa}_{t,\perp}$ satisfies $\mathbf{k}_t^\top \boldsymbol{\kappa}_{t,\perp} = 0$ and leaves Equation (44) unchanged while still perturbing the readout at other keys. The same-token residual is therefore blind to exactly the degree of freedom that distinguishes DiagKLA from IsoKLA, which is what Equation (110) is designed to isolate.

To expose causal ordering, temporarily allow a token-specific scale s_t . Current DiagKLA first computes $\hat{\mathbf{p}}_t$ and $\boldsymbol{\kappa}_t$ from the incoming state $\mathbf{c}_{t-1}$, and only afterward computes the tracked projected covariance below. In this subsection, plain $\mathbf{p}_t$ denotes that DiagKLA state, not the exact posterior diagonal.

$$p_{t,i}(s_t) = \left(\hat{p}_{t,i}^{-1} + s_t \frac{k_{t,i}^2}{r_t} \right)^{-1} = \frac{\hat{p}_{t,i}}{1 + s_t \hat{p}_{t,i} k_{t,i}^2 / r_t}. \quad (45)$$

Consequently, for a local intervention that holds the incoming state and token projections fixed,

$$\frac{\partial \boldsymbol{\kappa}_t}{\partial s_t} = 0, \quad \frac{\partial \mathbf{S}_t}{\partial s_t} = 0, \quad \frac{\partial p_{t,i}}{\partial s_t} = -\frac{k_{t,i}^2}{r_t} p_{t,i}^2 \leq 0. \quad (46)$$

The scale at token t therefore does not write that observation harder or change its same-token residual. It only increases the confidence state supplied to the future. In causal order, $\mathbf{c}_{t-1}$ determines $\hat{\mathbf{p}}_t$, which determines $\boldsymbol{\kappa}_t$ and $\mathbf{S}_t$; s_t then enters $\mathbf{c}_t$ and can first affect the memory gain at token $t+1$.

This token-local statement must not be confused with a comparison between two complete runs that use different global values of s . Except at the first token, a global s has already changed $\mathbf{c}_{t-1}$ and therefore generally changes the current gain. Conditional on fixed projected $\mathbf{k}$, $\boldsymbol{\alpha}$, $\boldsymbol{\omega}$, and r , its first delayed effect is explicit:

$$\frac{\partial z_{t+1}}{\partial s_t} = -\frac{1}{r_t} \sum_i \alpha_{t+1,i}^2 p_{t,i}^2 k_{t,i}^2 k_{t+1,i}^2 \leq 0, \quad (47)$$

$$\frac{\partial \beta_{t+1}^{\text{eff}}}{\partial s_t} = \frac{r_{t+1}}{(r_{t+1} + z_{t+1})^2} \frac{\partial z_{t+1}}{\partial s_t} \leq 0. \quad (48)$$

More generally, for $u > t$ the channelwise sensitivity propagates as

$$\frac{\partial p_{u,i}}{\partial s_t} = -\frac{k_{t,i}^2}{r_t} p_{t,i}^2 \prod_{n=t+1}^u \frac{\alpha_{n,i}^2}{(1 + s_n \hat{p}_{n,i} k_{n,i}^2 / r_n)^2}. \quad (49)$$

A larger past scale therefore reduces the later effective correction $\mathbf{k}_u^\top \boldsymbol{\kappa}_u$. It need not reduce every component or the norm of an anisotropic $\boldsymbol{\kappa}_u$: reducing the numerator coordinates also reduces their shared denominator. End-to-end trained runs need not be monotone either, because their learned projections can compensate for the changed recurrence.

Equations (48)–(49) also show what is being protected. Suppression is controlled by elementwise-energy overlap such as $\sum_i p_{t,i}^2 k_{t,i}^2 k_{u,i}^2$, not by the signed directional overlap $(\mathbf{k}_t^\top \mathbf{k}_u)^2$. At fixed

projections, a future key with disjoint coordinate support receives no direct effective-gain suppression through this pathway, although the full memory can still differ through intervening writes. Dense orthogonal keys can instead have large energy overlap, including the maximum compatible with orthogonality in the two-dimensional example above. The tracker gives those two keys identical confidence increments and, under a symmetric incoming covariance, identical effective-gain suppression. Their signed memory updates and residuals need not be identical. Large s can consequently suppress a genuinely new orthogonal association rather than selectively preserve only the old one.

Prediction limits the duration of this protection. As $s_t \rightarrow \infty$, $p_{t,i} \rightarrow 0$ wherever $k_{t,i} \neq 0$, but

$$\hat{p}_{t+1,i} = \alpha_{t+1,i}^2 p_{t,i} + \omega_{t+1,i} \longrightarrow \omega_{t+1,i}. \quad (50)$$

Positive process noise restores a floor on uncertainty and hence future plasticity; small α also rapidly attenuates the confidence sensitivity in Equation (49) while directly decaying the memory. Thus larger s does not store more content or create capacity by itself. It changes the retention–plasticity operating point until ω and α wash out that change. If $\omega = 0$ and $\alpha \simeq 1$, an extreme s can instead freeze future gains on the affected coordinates.

The different scale sensitivities of DiagKLA and IsoKLA are consistent with their different recurrences. For a spread-out normalized key, the DiagKLA increment in one coordinate has expectation $s/(d_k r_t)$, so $s = d_k$ cancels the coordinate dilution without changing the current DiagKLA gain locally. IsoKLA has one scalar state and $\|\mathbf{k}_t\|_2^2 = 1$, so its information increment is already s/r_t with no $1/d_k$ dilution. Moreover, the current IsoKLA implementation uses, locally holding the incoming $\hat{b}_t$ fixed,

$$\beta_t^{\text{iso}}(s) = \frac{\hat{b}_t}{r_t + s\hat{b}_t}, \quad c_t^{\text{iso}}(s) = \hat{b}_t^{-1} + \frac{s}{r_t}, \quad \frac{\partial \beta_t^{\text{iso}}}{\partial s} \bigg|_{\hat{b}_t} = -\frac{\hat{b}_t^2}{(r_t + s\hat{b}_t)^2} < 0. \quad (51)$$

A token-local increase in s in IsoKLA therefore weakens the current write and increases future confidence at the same time. With fixed projections, the lower uncertainty inherited from past larger scales reinforces this reduction in the global- s recurrence. At $s = d_k$ these two anti-plasticity effects can be excessive, whereas DiagKLA obtains delayed protection without the same local write penalty. This gives a mechanistic interpretation of why the production defaults $s = d_k$ and $s = 1$ behave differently. The corrected 220M sweeps do not establish unique optima: general quality is nearly flat across scales, while larger s trades toward stronger long-context retention. This is not yet causal evidence that retention alone produces the downstream differences.

A coherent Gaussian observation-precision multiplier has a different meaning. For a local intervention that holds the predicted prior $\hat{\mathbf{P}}_t$ fixed, increasing precision by s means using $r_t^{\text{eff}} = r_t/s$ in both the current gain and posterior. With $z_t = \mathbf{k}_t^\top \hat{\mathbf{P}}_t \mathbf{k}_t$, this gives

$$\kappa_t^{\text{coh}}(s) = \frac{s\hat{\mathbf{P}}_t \mathbf{k}_t}{r_t + sz_t}, \quad \beta_t^{\text{coh}}(s) = \frac{sz_t}{r_t + sz_t}, \quad \frac{\partial \beta_t^{\text{coh}}}{\partial s} = \frac{r_t z_t}{(r_t + sz_t)^2} > 0. \quad (52)$$

The remaining current residual is then $1 - \beta_t^{\text{coh}} = r_t/(r_t + sz_t)$, so a more precise observation is assimilated more aggressively. At the same time, its exact posterior is

$$\mathbf{P}_t(s) = \hat{\mathbf{P}}_t - \frac{s\hat{\mathbf{P}}_t \mathbf{k}_t \mathbf{k}_t^\top \hat{\mathbf{P}}_t}{r_t + sz_t}, \quad \frac{\partial \ell^\top \mathbf{P}_t(s) \ell}{\partial s} = -\frac{r_t (\ell^\top \hat{\mathbf{P}}_t \mathbf{k}_t)^2}{(r_t + sz_t)^2} \leq 0. \quad (53)$$

Thus coherent higher precision has the two temporally distinct effects expected of a Kalman filter: write the observation harder now, then reduce later corrections in covariance-overlapping directions. Neither the current DiagKLA split nor Equation (51) for $s \neq 1$ has this interpretation. A coherent

implementation should use r_t/s once; replacing r_t by r_t/s and also retaining a separate factor s in the information update would double-count the scale and produce an s^2 increment.

These mechanisms can be separated explicitly with a current-write scale s_{write} and a post-write confidence scale s_{protect} :

$$\kappa_t(s_{\text{write}}) = \frac{s_{\text{write}} \hat{\mathbf{P}}_t \odot \mathbf{k}_t}{r_t + s_{\text{write}} z_t}, \quad c_{t,i} = \hat{p}_{t,i}^{-1} + s_{\text{protect}} \frac{k_{t,i}^2}{r_t}. \quad (54)$$

Current DiagKLA is $(s_{\text{write}}, s_{\text{protect}}) = (1, s)$. Setting both to s makes the scale consistent within the diagonal precision projection, although that projection still is not the exact posterior diagonal.

A split variational objective. The two-scale rule is the stationary point of a single generalized variational free energy. For the mean-field family in Equation (22), write $q_{t,j} = \mathcal{N}(\boldsymbol{\mu}_{t,j}, \text{diag}(\mathbf{p}_t))$ and let the predicted prior be $\hat{\pi}_{t,j} = \mathcal{N}(\hat{\mathbf{s}}_{t,j}, \hat{\mathbf{P}}_t)$. Define

$$\begin{aligned} \mathcal{J}_{t,j}^{\text{split}}(q_{t,j}; s_{\text{write}}, s_{\text{protect}}) &= \text{KL}(q_{t,j} \parallel \hat{\pi}_{t,j}) + \frac{s_{\text{write}}}{2r_t} \left(v_{t,j} - \mathbf{k}_t^\top \boldsymbol{\mu}_{t,j} \right)^2 \\ &\quad + \frac{s_{\text{protect}}}{2r_t} \mathbf{k}_t^\top \text{diag}(\mathbf{p}_t) \mathbf{k}_t. \end{aligned} \quad (55)$$

The first data term penalizes error at the posterior mean and therefore controls current assimilation. The second penalizes predictive variance and therefore controls how much confidence is passed to later tokens. For diagonal $\hat{\mathbf{P}}_t = \text{diag}(\hat{\mathbf{p}}_t)$, differentiating Equation (55) gives the unique optimum

$$\boldsymbol{\mu}_{t,j}^* = \hat{\mathbf{s}}_{t,j} + \frac{s_{\text{write}} \hat{\mathbf{P}}_t \mathbf{k}_t}{r_t + s_{\text{write}} z_t} \left(v_{t,j} - \mathbf{k}_t^\top \hat{\mathbf{s}}_{t,j} \right), \quad (56)$$

$$(p_{t,i}^*)^{-1} = \hat{p}_{t,i}^{-1} + s_{\text{protect}} \frac{k_{t,i}^2}{r_t}. \quad (57)$$

The mean coefficient in Equation (56) is exactly the gain in Equation (54), and Equation (57) is its protection update. Current DiagKLA is therefore the optimizer of $\mathcal{J}^{\text{split}}(q_{t,j}; 1, s)$.

The connection to the power-posterior ELBO in Equation (39) follows by expanding its expected Gaussian negative log likelihood. Define

$$R_{t,j} = \left(v_{t,j} - \mathbf{k}_t^\top \boldsymbol{\mu}_{t,j}^* \right)^2, \quad V_t = \mathbf{k}_t^\top \text{diag}(\mathbf{p}_t) \mathbf{k}_t. \quad (58)$$

Then

$$\mathbb{E}_{q_{t,j}} \left[\left(v_{t,j} - \mathbf{k}_t^\top \boldsymbol{\mu}_{t,j}^* \right)^2 \right] = R_{t,j} + V_t. \quad (59)$$

Consequently, the negative power ELBO is

$$-\mathcal{L}_{t,j}^{(s)}(q_{t,j}) = \text{KL}(q_{t,j} \parallel \hat{\pi}_{t,j}) + \frac{s}{2r_t} (R_{t,j} + V_t) + \frac{s}{2} \log(2\pi r_t). \quad (60)$$

Remove only the Gaussian normalizer, which is independent of $q_{t,j}$ for fixed r_t , by defining

$$\bar{\mathcal{J}}_{t,j}^{(s)}(q_{t,j}) = -\mathcal{L}_{t,j}^{(s)}(q_{t,j}) - \frac{s}{2} \log(2\pi r_t). \quad (61)$$

Equation (55) can then be written exactly as

$$\mathcal{J}_{t,j}^{\text{split}}(q_{t,j}; s_{\text{write}}, s_{\text{protect}}) = \bar{\mathcal{J}}_{t,j}^{(s_{\text{write}})}(q_{t,j}) + \frac{s_{\text{protect}} - s_{\text{write}}}{2r_t} V_t. \quad (62)$$

Thus equal scales recover the power-posterior variational objective up to its q -independent normalizer. Current DiagKLA has the particularly simple form

$$\mathcal{J}_{t,j}^{\text{split}}(q_{t,j}; 1, s) = \overline{\mathcal{J}}_{t,j}^{(1)}(q_{t,j}) + \frac{s-1}{2r_t} V_t. \quad (63)$$

It is ordinary mean-field variational inference plus an additional predictive-variance penalty: the extra term increases future protection without strengthening the current mean update. A single Gaussian likelihood must weight $R_{t,j}$ and V_t equally, so unequal scales do not define the ELBO of one coherent observation model. They instead define a convex, decision-oriented mean-uncertainty objective with separate costs for present assimilation and future protection. If r_t is optimized through this objective rather than held fixed during the variational update, its Gaussian normalizer in Equation (60) must be retained.

A factorial comparison of $(1, 1)$, $(1, d_k)$, $(d_k, 1)$, and (d_k, d_k) would separate current assimilation from future protection; because $s_{\text{write}} = d_k$ may saturate the current correction, a gain-matched intermediate value should also be included. Log the same-token β_t^{eff} and residual contraction, the next-token and repeated-key gains, write/erase magnitudes, covariance quantiles, and effective memory half-life. Repeated-key, replacement, and orthogonal-key tasks can then identify whether the benefit comes from retention, impaired rewriting, or diagonal geometry.

A large learned r_t could make the $s = 1$ confidence increment too weak, but the preferred $s = d_k$ does not by itself diagnose observation noise as too large: increasing r_t also weakens the current gain, whereas increasing only s_{protect} does not. The relevant dimensionless quantities are z_t/r_t for the current write and $s\hat{p}_{t,i}k_{t,i}^2/r_t$ for the confidence update. These, together with $\omega_{t,i}/\hat{p}_{t,i}$, should be measured before attributing the result to r_t . More generally, if r_t is freely learned, the triplet $(r_t, s_{\text{write}}, s_{\text{protect}})$ is overparameterized: the updates depend only on

$$r_t^{\text{write}} = \frac{r_t}{s_{\text{write}}}, \quad \rho_t = \frac{s_{\text{protect}}}{s_{\text{write}}}. \quad (64)$$

Here r_t^{write} sets current assimilation and ρ_t is the identifiable protection-to-write ratio. The present decoupled scale is not redundant: current DiagKLA fixes $s_{\text{write}} = 1$ and uses $\rho_t = s$ to change future protection without changing that observation’s same-token write locally.

6.5 An exact posterior-diagonal alternative

Although the full posterior is dense, its diagonal after one measurement can be computed without materializing a $d_k \times d_k$ covariance. Define

$$D_t = r_t + \sum_j \hat{p}_{t,j} k_{t,j}^2, \quad q_{t,i} = \hat{p}_{t,i} k_{t,i}^2. \quad (65)$$

The denominator D_t is already required for the diagonal Kalman gain. The exact posterior diagonal from Equation (14) can therefore be updated as

$$p_{t,i}^{\text{mom}} = \hat{p}_{t,i} \frac{D_t - q_{t,i}}{D_t} = \hat{p}_{t,i} (1 - \kappa_{t,i} k_{t,i}), \quad \kappa_{t,i} = \frac{\hat{p}_{t,i} k_{t,i}}{D_t}. \quad (66)$$

This costs $O(d_k)$ state and arithmetic per token. The multiplicative form in Equation (66) is preferable to subtracting two nearly equal covariance terms. Since $D_t - q_{t,i} = r_t + \sum_{j \neq i} q_{t,j} > 0$, it also makes positivity explicit. Equivalently, the marginal information update is

$$\frac{1}{p_{t,i}^{\text{mom}}} = \frac{1}{\hat{p}_{t,i}} + \frac{k_{t,i}^2}{r_t + \sum_{j \neq i} \hat{p}_{t,j} k_{t,j}^2}. \quad (67)$$

Unlike the current update, this expression retains the competition from the other key coordinates instead of replacing its denominator by r_t .

The word “exact” applies only to the diagonal of this one posterior when the predicted prior is diagonal. The same update creates off-diagonal entries

$$P_{t,ij} = -\frac{\hat{p}_{t,i}\hat{p}_{t,j}k_{t,i}k_{t,j}}{D_t}, \quad i \neq j. \quad (68)$$

These entries affect the next gain. Discarding them and carrying only $\mathbf{p}_t^{\text{mom}}$ is therefore a moment-projected, diagonal assumed-density filter, not the exact multi-token Kalman filter. Exact filtering across tokens requires the dense $\mathbf{P}_t$, with $O(d_k^2)$ state and arithmetic per token, as in ExactKLA.

This alternative also changes the parallelization structure. All channels in Equation (66) are coupled through D_t , so the recurrence does not decompose into the independent 2×2 Möbius maps used by the current gain scan. The straightforward implementation is a token-sequential $O(Td_k)$ gain recurrence, preferably fused in Triton; after the gains are produced, the existing chunked Kalman memory kernel remains applicable. Thus the exact posterior-diagonal update saves the dense $O(d_k^2)$ covariance but trades away the current logarithmic-depth prefix scan for the gain.

For the moment projection, the coherent scale from Subsection 6.4 specializes to $r_t^{\text{eff}} = r_t/s$ and gives

$$D_t^{(s)} = \frac{r_t}{s} + \sum_j \hat{p}_{t,j}k_{t,j}^2, \quad \kappa_{t,i}^{(s)} = \frac{\hat{p}_{t,i}k_{t,i}}{D_t^{(s)}}, \quad p_{t,i}^{(s)} = \hat{p}_{t,i} - \frac{\hat{p}_{t,i}^2k_{t,i}^2}{D_t^{(s)}}. \quad (69)$$

This is the exact-posterior-diagonal specialization of Equations (52)–(53). A direct ablation should compare the current Möbius precision scan against this recurrent moment projection first with $s = 1$, and then, separately, with the coherent calibrated choice $r_t^{\text{eff}} = r_t/d_k$.

6.6 A tighter one-step approximation that preserves the scan

Yes, there is a better approximation in either of two relevant senses. If only one-step posterior fidelity matters, the moment projection above exactly matches the marginal variances. If the independent scan is a hard constraint, the process-noise-floor rule below is pointwise no worse than the $s = 1$ and $s = d_k$ rules, and generically better, in conditional one-step marginal-variance error at the cost of a token-local reduction.

Equation (67) is the forward-KL moment projection of this one-step posterior: it matches every exact marginal variance. Its obstacle is not arithmetic cost but scan structure, because the update for coordinate i depends on the recurrent uncertainty in every coordinate $j \neq i$. If the independent 2×2 Möbius scan is a hard constraint, a pointwise no-worse approximation can nevertheless retain the part of this cross-coordinate interference that is known before the scan, with strict improvement under the conditions below.

In particular, the prediction step and nonnegative process noise imply

$$\hat{p}_{t,j} = \alpha_{t,j}^2 p_{t-1,j} + \omega_{t,j} \geq \omega_{t,j}. \quad (70)$$

Define the token-local process-noise energy outside coordinate i by

$$L_t^\omega = \sum_j \omega_{t,j}k_{t,j}^2, \quad L_{t,-i}^\omega = L_t^\omega - \omega_{t,i}k_{t,i}^2, \quad (71)$$

and use the information increment

$$u_{t,i}^{\text{floor}} = \frac{k_{t,i}^2}{r_t + L_{t,-i}^\omega}, \quad p_{t,i}^{\text{floor}} = \left(\hat{p}_{t,i}^{-1} + u_{t,i}^{\text{floor}} \right)^{-1}. \quad (72)$$

Compared with the $s = 1$ reverse-KL rule, this update no longer pretends that the other key coordinates have zero uncertainty: it accounts for the uncertainty injected at the current prediction step. It becomes the exact moment projection whenever $\hat{p}_{t,j} = \omega_{t,j}$ for every active $j \neq i$, for example when there is no carried uncertainty in those coordinates.

The improvement is pointwise, not merely heuristic. Let

$$S_{t,-i} = \sum_{j \neq i} \hat{p}_{t,j} k_{t,j}^2, \quad u_{t,i}^{\text{mom}} = \frac{k_{t,i}^2}{r_t + S_{t,-i}}, \quad u_{t,i}^{\text{rev}} = \frac{k_{t,i}^2}{r_t}. \quad (73)$$

Equations (70)–(71) give $0 \leq L_{t,-i}^\omega \leq S_{t,-i}$. Hence, for $d_k \geq 1$,

$$d_k \frac{k_{t,i}^2}{r_t} \geq u_{t,i}^{\text{rev}} \geq u_{t,i}^{\text{floor}} \geq u_{t,i}^{\text{mom}}, \quad p_{t,i}^{\text{proj}, d_k} \leq p_{t,i}^{\text{rev}} \leq p_{t,i}^{\text{floor}} \leq p_{t,i}^{\text{mom}}. \quad (74)$$

Thus, holding the incoming $\hat{\mathbf{p}}_t$ fixed, $p_{t,i}^{\text{floor}}$ is always at least as close in absolute variance to the exact one-step marginal as either the $s = 1$ rule or the production $s = d_k$ rule. The improvement over $s = 1$ is strict when $k_{t,i} \neq 0$ and $L_{t,-i}^\omega > 0$. This ordering concerns posterior fidelity only; it does not imply that the less confident state is a better associative-memory controller, nor does it directly order the trajectories produced after the different states are fed into later predictions.

Crucially, L_t^ω is a current-token reduction: it depends only on $(\omega_t, \mathbf{k}_t)$ and not on a recurrent state. Substituting $u_{t,i}^{\text{floor}}$ for $u_{t,i}$ in Equation (35) gives

$$\mathbf{M}_{t,i}^{\text{floor}} = \begin{pmatrix} 1 + u_{t,i}^{\text{floor}} \omega_{t,i} & u_{t,i}^{\text{floor}} \alpha_{t,i}^2 \\ \omega_{t,i} & \alpha_{t,i}^2 \end{pmatrix}. \quad (75)$$

The mathematical prefix-scan and carry structure is therefore unchanged. An implementation still has to modify the token-map builder and its backward pass: it must either store L_t^ω or recompute one $O(d_k)$ cross-coordinate reduction wherever token maps are reconstructed, and the backward acquires coupled gradients through that reduction. In finite precision, $L_{t,-i}^\omega$ should be evaluated as $\max\{L_t^\omega - \omega_{t,i} k_{t,i}^2, 0\}$ to preserve the analytic nonnegativity when a key is nearly one-hot.

There is also a sharp reason not to expect an exact marginal update from any independent input-only scan. Holding the current-token projections fixed, for $j \neq i$ Equation (67) has

$$\frac{\partial (p_{t,i}^{\text{mom}})^{-1}}{\partial p_{t-1,j}} = -\frac{k_{t,i}^2 \alpha_{t,j}^2 k_{t,j}^2}{(r_t + S_{t,-i})^2}, \quad (76)$$

which is nonzero precisely when $k_{t,i} k_{t,j} \alpha_{t,j} \neq 0$. By contrast, a per-coordinate map $c_{t,i} = F_{t,i}(c_{t-1,i}; \mathbf{x}_t)$ whose coefficients depend only on the current token has zero derivative with respect to $p_{t-1,j}$. Therefore no single independent raw-input 2×2 Möbius scan can reproduce the exact transition for arbitrary prior diagonal states when $k_{t,i} k_{t,j} \alpha_{t,j} \neq 0$. Equation (72) is a principled compromise: it uses the simple, universally valid current-step lower bound on the omitted interference that requires no recurrent state, while preserving the production scan.

For direct fidelity to Equation (14), the rule should use r_t as written and no d_k multiplier. If stronger protection is useful for the mixer, introduce it explicitly rather than calling it a covariance approximation,

$$c_{t,i} = \hat{p}_{t,i}^{-1} + g_{\text{protect}} u_{t,i}^{\text{floor}}. \quad (77)$$

Here $g_{\text{protect}} = 1$ is the posterior-fidelity candidate, while $g_{\text{protect}} > 1$ is a retention calibration analogous to the current s_{protect} . To preserve the input-only scan, g_{protect} must be constant or a function only of the current token, not of a recurrent statistic. Once $g_{\text{protect}} \neq 1$, the ordering in Equation (74) need not hold.

6.7 Posterior fidelity is not mixer quality

The moment projection is exact for the one-step posterior diagonal, and the process-floor rule is a tighter conditional one-step marginal approximation than the current rule, but neither is necessarily a better drop-in strategy for associative memory. Consider the symmetric case $\hat{p}_{t,i} = \hat{p}_t$ and $k_{t,i}^2 = 1/d_k$, and define the effective write strength

$$\beta_t = \frac{\hat{p}_t}{r_t + \hat{p}_t}. \quad (78)$$

The current $s = d_k$ precision projection and the moment projection then give

$$p_t^{\text{proj}, d_k} = \hat{p}_t(1 - \beta_t), \quad p_t^{\text{mom}} = \hat{p}_t \left(1 - \frac{\beta_t}{d_k}\right). \quad (79)$$

One scalar measurement correctly removes only rank-one uncertainty, so the moment projection reduces each marginal variance by only a β_t/d_k fraction. The current rule reduces every coordinate by a β_t fraction, or d_k times as much total trace in this symmetric example. It therefore acts as a strong confidence regularizer: future gains are smaller and existing associations are overwritten less aggressively. Indeed, the current $s = 1$ projection also gives smaller posterior variance than the moment projection,

$$p_t^{\text{proj}, 1} = \frac{\hat{p}_t d_k r_t}{d_k r_t + \hat{p}_t} \leq p_t^{\text{mom}}, \quad (80)$$

with strict inequality for $d_k > 1$. The corrected information-scale sweep found standard-suite quality nearly flat and some long-context recall improved at larger s ; it did not establish d_k as a unique optimum. For $d_k > 1$, even the $s = 1$ projection retains less variance than the moment projection, so an unretuned moment variant is likely to produce larger future write/erase gains and more interference. It may nevertheless help tasks that require rapid adaptation or frequent replacement of old associations. Posterior faithfulness and memory quality are therefore distinct hypotheses.

The process-noise scale makes a raw comparison especially unfavorable to the moment variant. Prediction injects $O(d_k \omega_t)$ total diagonal uncertainty per token, whereas a rank-one moment update removes only $O(1)$ total uncertainty. Near $\alpha_t^2 = 1$, matching the stationary effective gain of the current $s = d_k$ rule requires approximately, when the retention-loss term proportional to $1 - \alpha_t^2$ is negligible,

$$\omega_t^{\text{mom}} \simeq \frac{\omega_t^{\text{proj}}}{d_k}, \quad (81)$$

or, at fixed process-noise scale and matching the stationary covariance-to-noise ratio, an observation-noise scale roughly d_k times larger. Without this recalibration, the current initialization $\omega_t \simeq \omega_{\min} + \text{softplus}(0)$ can drive the moment filter toward high uncertainty and β_t^{eff} close to one.

The appropriate comparison is therefore seven-way:

1. the production Möbius projection with $s = d_k$;
2. the same projection with $s = 1$;
3. the process-floor approximation with $g_{\text{protect}} = 1$;
4. the process-floor approximation with a fixed g_{protect} chosen offline to match the production next-token gain or memory-half-life distribution;
5. the recurrent moment projection with the same ω/r parameterization and $r_t^{\text{eff}} = r_t$;

6. the recurrent moment projection with the coherent likelihood scale $r_t^{\text{eff}} = r_t/d_k$ in both its same-token gain and posterior update;
7. the recurrent moment projection with ω initialized and floored at approximately $1/d_k$ of the production scale, or otherwise calibrated to match the initial and stationary β_t^{eff} distribution.

The raw comparisons measure the behavior of each formula under identical parameters; the gain- or half-life-matched comparisons isolate posterior geometry from overwrite temperature. All variants should log β_t^{eff} , covariance and process-noise quantiles, effective memory half-life, and write/erase magnitude. Repeated-key and orthogonal-key synthetic tasks should precede language model pretraining, because they directly separate rapid rewriting from protection against cross-talk. Among covariance approximations, the process-floor rule is the first candidate to test because it improves the conditional one-step marginal approximation without losing the parallel Möbius gain scan. The moment projection should replace it only if the matched comparison justifies the additional recurrent gain computation.

6.8 Anisotropic write direction and initialization

For a unit key, decompose

$$\beta_t^{\text{eff}} = \mathbf{k}_t^\top \boldsymbol{\kappa}_t, \quad \boldsymbol{\kappa}_{t,\perp} = \boldsymbol{\kappa}_t - \beta_t^{\text{eff}} \mathbf{k}_t, \quad \mathbf{k}_t^\top \boldsymbol{\kappa}_{t,\perp} = 0. \quad (82)$$

IsoKLA enforces $\boldsymbol{\kappa}_{t,\perp} = 0$: uncertainty changes the strength of a write while preserving the binding direction. DiagKLA can rotate both the write and erase directions away from the current key. This can be useful uncertainty preconditioning, or it can write the value into unrelated key directions and create cross-talk. Greater covariance expressivity alone does not distinguish these interpretations.

The current initialization is also not an exact IsoKLA fallback. Zeroing the output map of the process-noise projection makes $\omega_{t,i}$ uniform, but

$$\hat{p}_{t,i} = \alpha_{t,i}^2 / \mu + \omega_t \quad (83)$$

is already channel dependent when $\alpha_{t,i}$ is channel dependent. After one observation, $d_k k_{t,i}^2 / r_t$ further leaves the isotropic subspace. The initialization is a useful stabilizer, but the model cannot choose an exact IsoKLA function and add anisotropy only when it is beneficial.

6.9 Value-wise isotropic uncertainty

The columnwise Gaussian model in the main paper assigns the same key-space covariance to every value coordinate. In particular,

$$\begin{aligned} \tilde{\mathbf{s}}_{t,j} \mid \mathcal{F}_t &\sim \mathcal{N}(\mathbf{s}_{t,j}, \mathbf{P}_t) \quad \text{independently for } j = 1, \dots, d_v, \\ \text{Cov}\left(\text{vec}(\tilde{\mathbf{S}}_t) \mid \mathcal{F}_t\right) &= \mathbf{I}_{d_v} \otimes \mathbf{P}_t. \end{aligned} \quad (84)$$

Here and throughout this subsection, a tilde denotes the latent random memory, a bare symbol denotes its tracked mean, and a hat denotes a one-step prediction. A complementary structured approximation retains a different scalar key-space variance for each value coordinate:

$$\begin{aligned} \tilde{\mathbf{s}}_{t,j} \mid \mathcal{F}_t &\sim \mathcal{N}(\mathbf{s}_{t,j}, b_{t,j} \mathbf{I}_{d_k}) \quad \text{independently across } j, \\ \mathbf{B}_t &= \text{diag}(\mathbf{b}_t), \\ \text{Cov}\left(\text{vec}(\tilde{\mathbf{S}}_t) \mid \mathcal{F}_t\right) &= \mathbf{B}_t \otimes \mathbf{I}_{d_k}. \end{aligned} \quad (85)$$

The subscript j is essential: using the same $b_t \mathbf{I}$ for every column recovers Isotropic KDN rather than a new model. Equation (85) still assumes cross-column independence, but allows different value coordinates to have different reliability.

State-space model and projected prediction. To make the different column variances non-trivial, allow the process and observation uncertainty to depend on the value coordinate:

$$\begin{aligned}\tilde{\mathbf{s}}_{t,j} &= \mathbf{D}_t \tilde{\mathbf{s}}_{t-1,j} + \mathbf{w}_{t,j}, & \mathbf{w}_{t,j} &\sim \mathcal{N}(\mathbf{0}, \omega_{t,j} \mathbf{I}_{d_k}), \\ v_{t,j} &= \mathbf{k}_t^\top \tilde{\mathbf{s}}_{t,j} + e_{t,j}, & e_{t,j} &\sim \mathcal{N}(0, r_{t,j}), & \mathbf{D}_t &= \text{diag}(\boldsymbol{\alpha}_t).\end{aligned}\tag{86}$$

The noises are independent across columns and time. The current observation model is recovered by setting $r_{t,j} = r_t$, and shared process uncertainty is recovered by setting $\omega_{t,j} = \omega_t$. A non-diagonal value-noise covariance would couple the columns after conditioning and leave the factorized family in Equation (85).

The mean prediction remains shared,

$$\hat{\mathbf{s}}_{t,j} = \mathbf{D}_t \mathbf{s}_{t-1,j},\tag{87}$$

whereas the exact covariance prediction for column j is

$$\hat{\mathbf{P}}_{t,j}^{\text{exact}} = b_{t-1,j} \mathbf{D}_t \mathbf{D}_t^\top + \omega_{t,j} \mathbf{I}_{d_k}.\tag{88}$$

Following Isotropic KDN, project this covariance to its average marginal variance. Defining

$$a_t = \frac{1}{d_k} \text{Tr}(\mathbf{D}_t \mathbf{D}_t^\top) = \frac{1}{d_k} \sum_{i=1}^{d_k} \alpha_{t,i}^2\tag{89}$$

gives

$$\hat{\mathbf{P}}_{t,j} = \hat{b}_{t,j} \mathbf{I}_{d_k}, \quad \hat{b}_{t,j} = a_t b_{t-1,j} + \omega_{t,j}.\tag{90}$$

This prediction is exact when $\mathbf{D}_t \mathbf{D}_t^\top = a_t \mathbf{I}_{d_k}$; otherwise it is the covariance-coordinate trace projection used by the shared isotropic model.

Columnwise Kalman correction. Conditioned on the projected prediction, define the scalar innovation

$$\delta_{t,j} = v_{t,j} - \mathbf{k}_t^\top \hat{\mathbf{s}}_{t,j}.\tag{91}$$

The predictive cross-covariance and innovation variance are

$$\text{Cov}(\tilde{\mathbf{s}}_{t,j}, v_{t,j} \mid \mathcal{F}_{t-1}, \mathbf{k}_t) = \hat{b}_{t,j} \mathbf{k}_t, \quad \text{Var}(v_{t,j} \mid \mathcal{F}_{t-1}, \mathbf{k}_t) = r_{t,j} + \hat{b}_{t,j} \|\mathbf{k}_t\|_2^2.\tag{92}$$

The Gaussian conditional-mean identity therefore gives a different Kalman gain for every value coordinate,

$$\boldsymbol{\kappa}_{t,j} = \beta_{t,j} \mathbf{k}_t, \quad \beta_{t,j} = \frac{\hat{b}_{t,j}}{r_{t,j} + \hat{b}_{t,j} \|\mathbf{k}_t\|_2^2}, \quad \mathbf{s}_{t,j} = \hat{\mathbf{s}}_{t,j} + \beta_{t,j} \mathbf{k}_t \delta_{t,j}.\tag{93}$$

Thus all columns write along the same key direction, but they need not use the same write strength. The residual remaining at that key is

$$v_{t,j} - \mathbf{k}_t^\top \mathbf{s}_{t,j} = \frac{r_{t,j}}{r_{t,j} + \hat{b}_{t,j} \|\mathbf{k}_t\|_2^2} \delta_{t,j},\tag{94}$$

so every coordinate remains contractive even though the contraction factor is value dependent.

The exact posterior covariance and information for column j are

$$\begin{aligned}\mathbf{P}_{t,j}^{\text{exact}} &= \widehat{b}_{t,j} \mathbf{I}_{d_k} - \frac{\widehat{b}_{t,j}^2 \mathbf{k}_t \mathbf{k}_t^\top}{r_{t,j} + \widehat{b}_{t,j} \|\mathbf{k}_t\|_2^2}, \\ \mathbf{C}_{t,j}^{\text{exact}} &= (\mathbf{P}_{t,j}^{\text{exact}})^{-1} = \widehat{c}_{t,j} \mathbf{I}_{d_k} + \frac{1}{r_{t,j}} \mathbf{k}_t \mathbf{k}_t^\top, \quad \widehat{c}_{t,j} = \widehat{b}_{t,j}^{-1}.\end{aligned}\tag{95}$$

The rank-one term leaves the isotropic family, so the posterior must again be projected before it can be carried to the next token.

Reverse-KL isotropic projection. Let $\pi_{t,j}$ be the exact Gaussian posterior in Equation (95) and consider

$$q_{t,j} = \mathcal{N}(\mathbf{m}_{t,j}, b_{t,j} \mathbf{I}_{d_k}).\tag{96}$$

Up to terms independent of $\mathbf{m}_{t,j}$ and $b_{t,j}$, the reverse KL divergence is

$$\text{KL}(q_{t,j} \parallel \pi_{t,j}) = \frac{1}{2} \left[(\mathbf{m}_{t,j} - \mathbf{s}_{t,j})^\top \mathbf{C}_{t,j}^{\text{exact}} (\mathbf{m}_{t,j} - \mathbf{s}_{t,j}) + b_{t,j} \text{Tr}(\mathbf{C}_{t,j}^{\text{exact}}) - d_k \log b_{t,j} \right] + \text{const.}\tag{97}$$

The first-order conditions give

$$\mathbf{m}_{t,j} = \mathbf{s}_{t,j}, \quad b_{t,j} = \frac{d_k}{\text{Tr}(\mathbf{C}_{t,j}^{\text{exact}})} = \left(\widehat{c}_{t,j} + \frac{\|\mathbf{k}_t\|_2^2}{d_k r_{t,j}} \right)^{-1}.\tag{98}$$

The projection therefore preserves the exact one-step posterior mean under the projected predictive prior and retains one scalar variance for each memory column. In information variables $c_{t,j} = b_{t,j}^{-1}$,

$$\widehat{c}_{t,j} = \frac{c_{t-1,j}}{a_t + \omega_{t,j} c_{t-1,j}}, \quad u_{t,j} = \frac{\|\mathbf{k}_t\|_2^2}{d_k r_{t,j}}, \quad c_{t,j} = \widehat{c}_{t,j} + u_{t,j}.\tag{99}$$

This is the ordinary reverse-KL solution. Multiplying $u_{t,j}$ by an additional information scale would be a separate calibration rather than the variational optimum for the observation model in Equation (86).

Associative uncertainty scan and memory update. Substituting the prediction into the information update yields

$$c_{t,j} = \frac{(1 + u_{t,j} \omega_{t,j}) c_{t-1,j} + u_{t,j} a_t}{\omega_{t,j} c_{t-1,j} + a_t}.\tag{100}$$

Hence each value coordinate has the Möbius matrix

$$\mathbf{M}_{t,j} = \begin{bmatrix} 1 + u_{t,j} \omega_{t,j} & u_{t,j} a_t \\ \omega_{t,j} & a_t \end{bmatrix}, \quad \begin{bmatrix} n_{t,j} \\ d_{t,j} \end{bmatrix} = \mathbf{M}_{t,j} \begin{bmatrix} n_{t-1,j} \\ d_{t-1,j} \end{bmatrix}, \quad c_{t,j} = \frac{n_{t,j}}{d_{t,j}}, \quad \begin{bmatrix} n_{0,j} \\ d_{0,j} \end{bmatrix} = \begin{bmatrix} c_{0,j} \\ 1 \end{bmatrix}.\tag{101}$$

Matrix multiplication therefore gives an associative gain scan, now comprising d_v independent scalar scans rather than one shared scalar scan.

Let $\boldsymbol{\beta}_t = (\beta_{t,1}, \dots, \beta_{t,d_v})^\top$ and $\widehat{\mathbf{S}}_t = \mathbf{D}_t \mathbf{S}_{t-1}$. Stacking the exact columnwise means from Equation (93) gives

$$\begin{aligned}\mathbf{S}_t &= \widehat{\mathbf{S}}_t + \mathbf{k}_t \left[\boldsymbol{\beta}_t \odot \left(\mathbf{v}_t - \widehat{\mathbf{S}}_t^\top \mathbf{k}_t \right) \right]^\top \\ &= \widehat{\mathbf{S}}_t - \mathbf{k}_t \mathbf{k}_t^\top \widehat{\mathbf{S}}_t \text{diag}(\boldsymbol{\beta}_t) + \mathbf{k}_t (\boldsymbol{\beta}_t \odot \mathbf{v}_t)^\top.\end{aligned}\tag{102}$$

6.9.1 Comparison with Gated DeltaNet-2

Using the same predicted memory $\hat{\mathbf{S}}_t = \mathbf{D}_t \mathbf{S}_{t-1}$, Gated DeltaNet-2 (GDN-2) can be written as

$$\begin{aligned} \mathbf{S}_t^{\text{GDN2}} &= \hat{\mathbf{S}}_t - \mathbf{k}_t (\mathbf{b}_t^{\text{erase}} \odot \mathbf{k}_t)^\top \hat{\mathbf{S}}_t + \mathbf{k}_t (\mathbf{w}_t \odot \mathbf{v}_t)^\top \\ &= \hat{\mathbf{S}}_t + \mathbf{k}_t \left[\mathbf{w}_t \odot \mathbf{v}_t - \hat{\mathbf{S}}_t^\top (\mathbf{b}_t^{\text{erase}} \odot \mathbf{k}_t) \right]^\top, \end{aligned} \quad (103)$$

where $\mathbf{b}_t^{\text{erase}} \in \mathbb{R}^{d_k}$ is a key-channel erase gate and $\mathbf{w}_t \in \mathbb{R}^{d_v}$ is a value-channel write gate. Both are predicted directly from the current token. Equation (102) and GDN-2 therefore share the outer write direction $\mathbf{k}_t$ and both permit value-wise scaling of the incoming value. Their erase terms, however, act on different axes:

$$\begin{aligned} \text{value-wise isotropic:} & \quad -\mathbf{k}_t \mathbf{k}_t^\top \hat{\mathbf{S}}_t \text{diag}(\beta_t), \\ \text{GDN-2:} & \quad -\mathbf{k}_t (\mathbf{b}_t^{\text{erase}} \odot \mathbf{k}_t)^\top \hat{\mathbf{S}}_t. \end{aligned} \quad (104)$$

In the value-wise isotropic filter, column j uses the same uncertainty-derived $\beta_{t,j}$ to scale both the observation write and the predicted-value erasure. GDN-2 instead uses one key-wise pseudo-key $\mathbf{b}_t^{\text{erase}} \odot \mathbf{k}_t$ to erase every value column, while the independent $w_{t,j}$ scales only the incoming value. Thus GDN-2 deliberately decouples the two terms that a Kalman innovation ties together.

The equality conditions make the relationship precise. For Equations (102) and (103) to agree for arbitrary predicted memories and values, we must have

$$w_{t,j} = \beta_{t,j}, \quad \mathbf{b}_t^{\text{erase}} \odot \mathbf{k}_t = \beta_{t,j} \mathbf{k}_t \quad \text{for every } j. \quad (105)$$

For a generic key, the second condition forces all $\beta_{t,j}$ to equal one scalar β_t and forces $b_{t,i}^{\text{erase}} = \beta_t$ on the support of $\mathbf{k}_t$. Together with the first condition,

$$\mathbf{b}_t^{\text{erase}} = \beta_t \mathbf{1}_{d_k}, \quad \mathbf{w}_t = \beta_t \mathbf{1}_{d_v}, \quad (106)$$

which is exactly the KDA recurrence; a scalar transition further recovers Gated DeltaNet. Consequently, neither general family contains the other. Their common intersection is the scalar-gated delta rule.

GDN-2 can still be written as a fixed-gain Kalman-like residual after defining the pseudo-key $\mathbf{e}_t = \mathbf{b}_t^{\text{erase}} \odot \mathbf{k}_t$, pseudo-value $\mathbf{z}_t = \mathbf{w}_t \odot \mathbf{v}_t$, and prescribing the gain $\mathbf{k}_t$. It is not, however, the posterior-mean update of the original observation model $\mathbf{v}_t = \tilde{\mathbf{S}}_t^\top \mathbf{k}_t + \text{noise}$ under a column-specific isotropic covariance. In that model the same gain must multiply both halves of each innovation.

The correction remains rank one, but it is no longer of the shared-gain form $\hat{\mathbf{S}}_t + \kappa_t \delta_t^\top$. Equivalently, column j has transition

$$\mathbf{s}_{t,j} = \underbrace{(\mathbf{I} - \beta_{t,j} \mathbf{k}_t \mathbf{k}_t^\top)}_{\mathbf{A}_{t,j}} \mathbf{D}_t \mathbf{s}_{t-1,j} + \beta_{t,j} \mathbf{k}_t v_{t,j}. \quad (107)$$

The information stage remains scan-compatible, and the sequential memory update still costs $O(d_k d_v)$ per token. However, $\mathbf{A}_{t,j}$ varies with j , so the single compact-WY system shared across all value columns in the current KDN kernel no longer applies directly. An exact parallel implementation requires value-wise batched systems, or a grouped-value approximation in which columns within each group share one scalar uncertainty and gain.

When the extension is genuinely different. If

$$b_{0,j} = b_0, \quad \omega_{t,j} = \omega_t, \quad r_{t,j} = r_t \quad \text{for every } j, \quad (108)$$

then induction through Equations (90) and (99) gives $b_{t,j} = b_t$ and $\beta_{t,j} = \beta_t$ for every column and token. The observed values cannot break this symmetry because Gaussian covariance updates do not depend on the realized $v_{t,j}$. At least one of the initial uncertainty, process noise, or observation noise must therefore vary across value coordinates for the extension to differ from Isotropic KDN.

If hardware constraints instead force one common gain, minimizing total conditional Frobenius risk over the restricted family $\mathbf{s}_{t,j} = \hat{\mathbf{s}}_{t,j} + \boldsymbol{\kappa}_t \delta_{t,j}$ gives

$$\boldsymbol{\kappa}_t^{\text{shared}} = \frac{\sum_{j=1}^{d_v} \hat{b}_{t,j}}{\sum_{j=1}^{d_v} r_{t,j} + \|\mathbf{k}_t\|_2^2 \sum_{j=1}^{d_v} \hat{b}_{t,j}} \mathbf{k}_t. \quad (109)$$

For shared r_t , this is the isotropic gain formed from $\bar{b}_t = d_v^{-1} \sum_j \hat{b}_{t,j}$. It is the best common-gain compromise, but it is not the columnwise Kalman posterior mean unless all $\beta_{t,j}$ coincide; enforcing it largely averages away the extra uncertainty state at the current write.

Allowing independent $\omega_{t,j}$ and $r_{t,j}$ also introduces one covariance-scale gauge per value coordinate: jointly scaling $b_{t,j}$, $\omega_{t,j}$, and $r_{t,j}$ leaves $\beta_{t,j}$ unchanged. A fixed or anchored observation-noise scale is therefore even more important in this model; tying $r_{t,j} = r_t$ reduces these independent gauges to the common gauge already discussed above.

This model and Diagonal KDN spend a similar scalar state budget on different axes. Diagonal KDN tracks key-channel uncertainty $p_{t,i}$ and produces one anisotropic gain shared by all value columns. The value-wise isotropic model tracks value-coordinate uncertainty $b_{t,j}$ and produces value-dependent scalar gains that always point along $\mathbf{k}_t$. It is principled when value coordinates have genuinely different volatility or observation reliability, but it does not represent the key-direction uncertainty responsible for associative interference. It also depends on the learned value basis and weakens the shared-kernel implementation. The cleanest use is therefore as a value-wise or grouped-value ablation rather than a drop-in replacement for Diagonal KDN.

6.10 Decisive ablation

The cleanest next experiment holds the diagonal information tracker and effective write strength fixed while intervening only on the gain direction:

$$\boldsymbol{\kappa}_t^{(\eta)} = \beta_t^{\text{eff}} \mathbf{k}_t + \eta \boldsymbol{\kappa}_{t,\perp}, \quad \eta \in \{0, 1\}. \quad (110)$$

Here $\eta = 0$ removes only the off-key component and $\eta = 1$ recovers the current DiagKLA. A gain for $\eta = 0$ would implicate the off-key direction; parity would show little measurable benefit from that direction; a robust gain for $\eta = 1$ would provide direct evidence that anisotropic writes add useful capacity. The comparison should log β^{eff} , gain angle, $\mathbf{p}$ and $\mathbf{c}$ quantiles, ω/r , and cold-channel fraction on current checkpoints.

Two supporting controls are needed:

1. Compare the projected information state in Equation (15), its $s = 1$ version, and the process-floor state in Equation (72) with both the sequential moment-projected oracle in Equation (14) and the dense ExactKLA covariance diagonal on a short synthetic task, including orthogonal keys with identical elementwise squares.

2. Run a matched three-seed 220M comparison of DiagKLA, IsoKLA, and the stabilized IsoKLA variant with identical world size, data order, and parameter count. Validate the S-NIAH-2 advantage across context lengths and needle depths before spending on a new large-scale run.

The immediate scale control is to complete the full long-context grid for the existing $s = d_k$ checkpoint and compare it with the completed $s = 4d_k$ arm; the currently comparable cells do not resolve their ordering. Only if $4d_k$ wins consistently is a point above that boundary warranted. This is an overwrite-temperature diagnostic: larger s increases the post-write confidence state. A token-local change does not strengthen that token’s write, while a global change alters later current writes through their covariance prefix. The sweep can locate a better retention–plasticity operating point, but it cannot resolve the posterior inconsistency or establish that off-key anisotropy is useful.

7 The Zero-Process-Noise Limit

Because ω carries part of the covariance-scale symmetry of Section 2 and is what makes the information prediction intractable in closed form, it is natural to ask whether the degenerate choice $\omega_t \equiv 0$ is preferable. It is not. The limit buys a cheaper scan, does not improve the fidelity of the diagonal approximation at all, and removes the plasticity floor that makes the layer a memory rather than an estimator.

7.1 It does not restore exactness

Prediction never creates off-diagonal covariance, at any ω . With $\mathbf{D}_t = \text{diag}(\alpha_t)$ and $\mathbf{\Omega}_t = \text{diag}(\omega_t)$,

$$\hat{P}_{t,ij} = \alpha_{t,i}\alpha_{t,j}P_{t-1,ij} + \omega_{t,i}\delta_{ij}, \quad (111)$$

so the predict step rescales the existing sparsity pattern and adds only to the diagonal. The measurement is the sole source of off-diagonal mass, and by Equation (68) it contributes $P_{t,ij} = -\hat{p}_{t,i}\hat{p}_{t,j}k_{t,i}k_{t,j}/D_t$, which vanishes for every $i \neq j$ if and only if $k_{t,i}k_{t,j} = 0$. The exactness condition for the diagonal family is therefore a condition on the keys alone,

$$\mathbf{P}_t \text{ diagonal for all } t \iff \text{every } \mathbf{k}_t \text{ is supported on a single coordinate}, \quad (112)$$

and process noise does not appear in it. Setting $\omega_t = 0$ leaves the approximation error of the diagonal state exactly where it was.

What the limit does make exact is the *information* prediction: for invertible $\mathbf{D}_t$ it becomes the conjugation $\hat{\mathbf{C}}_t = \mathbf{D}_t^{-\top} \mathbf{C}_{t-1} \mathbf{D}_t^{-1}$ rather than an inverse of a sum. That is a statement about the parameterization of an already-diagonal state, not about how faithfully that state represents the true posterior. The two senses of “exact” should not be conflated.

7.2 What the limit does buy

The general per-channel information map in Equation (34) is represented by the matrix in Equation (35); its 2×2 products admit a logarithmic-depth prefix scan. At $\omega_{t,i} = 0$ the lower-left entry vanishes, $\mathbf{M}_{t,i}$ becomes upper triangular, and the Möbius scan degenerates to the affine recurrence

$$c_{t,i} = \frac{c_{t-1,i}}{\alpha_{t,i}^2} + s \frac{k_{t,i}^2}{r_t}, \quad (113)$$

the ordinary decay-accumulate scan of gated linear attention. This is a real implementation saving, and it is the only advantage of the limit.

7.3 Covariance windup

A Kalman filter without process noise is recursive least squares, and it inherits the classical covariance windup of that estimator: information only accumulates, so the gain decays to zero and the layer stops writing. By Equation (44) a vanishing β_t^{eff} leaves the residual untouched, $\mathbf{v}_t - \mathbf{S}_t^\top \mathbf{k}_t \rightarrow \mathbf{e}_t$, so the write becomes a no-op rather than merely a weak one. There are two regimes.

- $\alpha < 1$. Measurement only decreases variance, so $p_{t,i} \leq \hat{p}_{t,i} = \alpha_{t,i}^2 p_{t-1,i}$ and hence $p_{t,i} \leq \alpha^{2t} p_{0,i}$. The gain dies geometrically while $\mathbf{D}_t$ is simultaneously draining $\mathbf{S}_t$: the memory empties and cannot be refilled. This is the degenerate corner.
- $\alpha = 1$. For spread-out unit keys, $\mathbb{E}[k_i^2] = 1/d_k$ gives $c_{t,i} = \mu + ts/(d_k r)$ and therefore

$$\beta_t^{\text{eff}} = \frac{d_k}{d_k(1 + \mu r) + st} \xrightarrow[t \text{ large}]{} \frac{d_k}{st}. \quad (114)$$

At the default $s = d_k$ this is $\beta_t^{\text{eff}} \simeq 1/t$, about 2×10^{-4} by token 4096. Larger s freezes the memory proportionally faster, with no floor to stop it.

7.4 The plasticity floor set by process noise

Positive process noise is precisely what prevents this. With $\alpha \simeq 1$, $s = d_k$, and spread-out unit keys, the tracked variance obeys $p_t = \hat{p}_t/(1 + \hat{p}_t/r)$ and $\hat{p}_{t+1} = p_t + \omega$, whose fixed point solves $\hat{p}_\infty^2 - \omega \hat{p}_\infty - \omega r = 0$:

$$\hat{p}_\infty = \frac{\omega + \sqrt{\omega^2 + 4\omega r}}{2} = \Theta(\sqrt{\omega r}) \quad (\omega \ll r), \quad \beta_\infty^{\text{eff}} = \frac{\hat{p}_\infty}{r + \hat{p}_\infty}. \quad (115)$$

The uncertainty floor is the geometric mean of process and observation noise. It therefore degrades gracefully rather than abruptly, but it vanishes continuously as $\omega \rightarrow 0$, recovering the windup above. Equation (115) is the stationary counterpart of Equation (50), which states the same protection only in the $s \rightarrow \infty$ limit at a single token. At the current initialization— $\omega_{\min} = r_{\min} = 0.05$ with zeroed output maps, giving $\omega \simeq r \simeq 0.74$ —it evaluates to $\hat{p}_\infty \simeq 1.20$ and $\beta_\infty^{\text{eff}} \simeq 0.62$, a healthy retention–plasticity operating point rather than a frozen one.

7.5 Two things the limit does not do

It does not remove the scale symmetry. Section 2 survives with one fewer parameter: $\mathbf{p}'_t = \lambda \mathbf{p}_t$, $r'_t = \lambda r_t$, $\mu' = \mu/\lambda$ still leaves $\boldsymbol{\kappa}_t$ invariant. The gauge choice $r \equiv 1$, $\mu \equiv 1$ is still required and is still the recommended fix.

It is also not the intervention that produced the noisy-MQAR improvement. That result came from zeroing the *output map* of the process-noise projection, which makes $\omega_{t,i}$ uniform across channels at $\omega_{\min} + \text{softplus}(0) \simeq 0.74$ —isotropic and large, not small. Setting $\boldsymbol{\omega} = 0$ is the opposite intervention, and conflating the two reads that result backwards. The parameterization $\omega_{t,i} = \omega_{\min} + \text{softplus}(f_\omega(\mathbf{x}_t))$ with $\omega_{\min} > 0$ makes the limit unreachable by design.

The zero-process-noise filter is the correct estimator only when the associations are fixed for the whole context and nothing must be overwritten, which is exactly the regime in which recursive least squares is optimal. For a streaming memory it is the degenerate corner that $\omega_{\min}$ exists to exclude.

8 Overwrite, Overconfidence, and Bounded Writes

The information-scale results expose a distinction that is easy to miss. Increasing the protection scale s records more confidence after a write and, by Equation (48), lowers the effective gains of later tokens. Better accuracy at larger s is therefore consistent with an overwrite-limited memory: stronger confidence protects existing associations from subsequent write-erase updates. It is not evidence of a better calibrated posterior. The reverse-KL state at $s = 1$ is already overconfident relative to the exact marginal by Equation (31), and $s > 1$ decreases its retained variance further. Statistical overconfidence and excessive overwrite are thus different, and often opposing, failure modes. The former suppresses future plasticity; the latter changes the current memory too aggressively.

8.1 Why a better diagonal projection cannot solve both problems

Avoiding posterior overconfidence and suppressing overwrite initially appear to ask for the same change, but a purely diagonal covariance exposes a conflict between them. Suppress the token index, let the predicted covariance be $\mathbf{P}^- = \text{diag}(p_1, \dots, p_{d_k})$, and define

$$q_i = p_i k_i^2, \quad z = \sum_i q_i, \quad D = r + z. \quad (116)$$

The exact one-step posterior, the reverse-KL diagonal projection, and the forward-KL moment projection are respectively

$$\mathbf{P}^* = \mathbf{P}^- - \frac{(\mathbf{P}^- \mathbf{k})(\mathbf{P}^- \mathbf{k})^\top}{D}, \quad (117)$$

$$p_i^{\text{rev}} = p_i - \frac{p_i^2 k_i^2}{r + q_i}, \quad p_i^{\text{mom}} = p_i - \frac{p_i^2 k_i^2}{D}. \quad (118)$$

Because $r + q_i \leq D$, the reverse-KL state satisfies $p_i^{\text{rev}} \leq p_i^{\text{mom}} = P_{ii}^*$. Replacing it by the moment projection therefore removes the one-step marginal overconfidence. It does not, however, protect the direction that was just observed.

To see why, consider the posterior variance left along the same key. With $f(x) = rx/(r+x)$,

$$\begin{aligned} v^* &:= \mathbf{k}^\top \mathbf{P}^* \mathbf{k} = f(z), \\ v^{\text{rev}} &:= \sum_i k_i^2 p_i^{\text{rev}} = \sum_i f(q_i), \\ v^{\text{mom}} &:= \sum_i k_i^2 p_i^{\text{mom}} = v^* + \frac{2 \sum_{i < j} q_i q_j}{D}. \end{aligned} \quad (119)$$

Since f is nonnegative and subadditive on $\mathbb{R}_+$,

$$v^* \leq v^{\text{rev}} \leq v^{\text{mom}}. \quad (120)$$

Thus the reverse-KL approximation can underestimate every marginal variance while still *overestimating* uncertainty in the observed direction. The missing quantity is the rank-one cross-coordinate correction in Equation (117), whose quadratic contribution along $\mathbf{k}$ is negative. It preserves uncertainty in individual coordinates while recording that a particular linear combination has already been measured. For an immediate repeated-key observation with noise r_{next} , the effective write $v/(r_{\text{next}} + v)$ is strictly increasing in v . The exact covariance can consequently be better calibrated and less willing to overwrite the same association at once. A diagonal moment projection keeps the correct marginals but discards precisely the correlations that provide this directional protection, so it can increase repeated-key writes relative to reverse KL.

Matching the exact repeated-key write with a scalar scale. If the immediate objective is only to match the protection in the direction that was just observed, the existing information scale provides a cheaper approximation than retaining the full covariance. Apply a scale s only to the diagonal precision passed forward, leaving the current mean gain unchanged:

$$p_i(s) = \left(p_i^{-1} + s \frac{k_i^2}{r} \right)^{-1} = \frac{r p_i}{r + s q_i}, \quad v(s) := \sum_i k_i^2 p_i(s) = \sum_i \frac{r q_i}{r + s q_i}. \quad (121)$$

The ordinary reverse-KL projection is $s = 1$, for which $v(1) = v^{\text{rev}} \geq v^*$. Hence an exact one-step directional calibration chooses s_{dir}^* to satisfy

$$\sum_i \frac{x_i}{1 + s_{\text{dir}}^* x_i} = \frac{X}{1 + X}, \quad x_i = \frac{q_i}{r}, \quad X = \sum_i x_i. \quad (122)$$

For $X > 0$, the left-hand side is continuous and strictly decreasing in s_{dir}^* , so the solution is unique. If d_{act} coordinates have $q_i > 0$, subadditivity at $s = 1$ and concavity at $s = d_{\text{act}}$ give $1 \leq s_{\text{dir}}^* \leq d_{\text{act}}$. Balanced contributions $q_i = q$ attain $s_{\text{dir}}^* = d_{\text{act}}$; in particular, a dense symmetric key recovers the width calibration $s = d_k$ derived earlier. In the low-SNR regime $z/r \ll 1$, expanding Equation (122) gives the participation-ratio approximation

$$s_{\text{dir}}^* \simeq d_{\text{eff}} := \frac{(\sum_i q_i)^2}{\sum_i q_i^2}. \quad (123)$$

Thus the empirically useful $s > 1$ can be interpreted as compensating, on average, for the directional confidence that was lost when the exact rank-one covariance correction was discarded.

By construction, Equation (122) makes $v(s_{\text{dir}}^*) = v^*$. With no intervening state dynamics or process noise, it therefore reproduces the exact effective gain $v^*/(r_{\text{next}} + v^*)$ for an immediate repeated key. This is only a directional scalar, not the full Kalman gain vector. This is only a directional match: it generally does not recover the exact marginal variances or gains for arbitrary future keys, and $s_{\text{dir}}^* > 1$ makes the coordinatewise marginal overconfidence of reverse KL more severe. Moreover, the exact root depends on the recurrent quantities $q_i = p_i k_i^2$. Computing it online couples the channels and prevents the current recurrence from being represented by input-only scalar Möbius maps. A fixed or headwise learned s , or an input-only proxy such as $\|\mathbf{k}\|_2^4 / \|\mathbf{k}\|_4^4$ when the predicted variances are nearly isotropic, preserves the existing associative scan but only amortizes the directional match. Restoring covariance geometry more generally requires a richer state family.

Block reverse-KL hierarchy. A structured compromise is to partition the key coordinates into blocks $\mathcal{G}_1, \dots, \mathcal{G}_m$ of size at most b and retain a dense covariance inside each block. For a block-diagonal predicted covariance, reverse-KL projection onto this family sets

$$\mathbf{P}_g^{\text{BR}} = \left[(\mathbf{P}_g^-)^{-1} + \frac{\mathbf{k}_g \mathbf{k}_g^\top}{r} \right]^{-1} = \mathbf{P}_g^- - \frac{\mathbf{P}_g^- \mathbf{k}_g \mathbf{k}_g^\top \mathbf{P}_g^-}{r + z_g}, \quad z_g = \mathbf{k}_g^\top \mathbf{P}_g^- \mathbf{k}_g. \quad (124)$$

The current posterior mean remains the exact Kalman mean under the block-diagonal predictive prior, with gain

$$\boldsymbol{\kappa}_g = \frac{\mathbf{P}_g^- \mathbf{k}_g}{r + \sum_{\ell=1}^m z_\ell}. \quad (125)$$

For the diagonal incoming covariance used in Equations (117)–(118), every $i \in \mathcal{G}_g$ obeys

$$p_i^{\text{rev}} \leq (\mathbf{P}_g^{\text{BR}})_{ii} \leq p_i^{\text{mom}}, \quad (126)$$

while the retained variance along the key is

$$v^{\text{BR}} = \sum_{g=1}^m f(z_g), \quad v^\star \leq v^{\text{BR}} \leq v^{\text{rev}} \leq v^{\text{mom}}. \quad (127)$$

Coarsening a nested block partition therefore moves the marginal variances upward toward their exact values while moving the repeated-key directional variance downward toward its exact value. Singleton blocks recover the diagonal reverse-KL update; one full block recovers the exact covariance. Unlike interpolation between forward and reverse diagonal KL, this hierarchy improves both quantities because it restores the missing correlations. These are one-step statements for a fixed predictive prior. They do not order arbitrary future keys, trajectories with intervening dynamics and process noise, realized updates with different residuals, or the loss after end-to-end training.

This improvement has a controlled but nontrivial cost. With fixed block size b , the covariance state uses $O(d_k b)$ numbers. Each block Riccati update can in principle be represented as a matrix-valued linear-fractional map and composed associatively, replacing each scalar 2×2 Möbius scan by a $2b \times 2b$ block scan when the block transitions are invertible and the transition and process covariance preserve the fixed partition. The current diagonal parameterization satisfies these structural conditions. A straightforward combine costs $O(d_k b^2)$ work across all blocks, and stable factorizations and backward kernels are substantially more involved than in the diagonal implementation. The sum $\sum_g z_g$ required by Equation (125) is token-local and does not couple the recurrent block maps. A diagonal-plus-low-rank covariance captures correlations across blocks more directly, but its rank grows with every observation; any fixed-rank truncation is nonlinear and order dependent, so exact associativity is lost.

For $d_k = 128$, a practical first hierarchy is the nested comparison $b \in \{1, 8, 16\}$, with $b = 128$ used only as the dense reference. The variants should be matched for their initial and stationary β_t^{eff} , or for measured memory half-life, so that the comparison isolates covariance geometry rather than a different write temperature. Diagnostics should include marginal-variance error against dense filtering, uncertainty in the most recent key direction, gains on repeated and unrelated keys, and downstream retrieval loss.

Block reverse KL is therefore the clean covariance-based candidate when a heavier scan is acceptable. Under the strict $O(d_k)$ scalar-scan budget, the process-noise-floor update in Subsection 6.6 remains the better fidelity approximation: it lies between reverse KL and moment matching and preserves the independent Möbius maps. Because it cannot retain posterior correlations, however, it does not guarantee less overwrite. An alternative is to replace covariance fidelity by fidelity to the filtering action itself.

8.2 From covariance projection to gain matching

The desired approximation can be posed directly before choosing a divergence between covariances. Define the gain operator

$$\mathbf{K}(\mathbf{P}; \mathbf{k}, r) := \frac{\mathbf{P}\mathbf{k}}{r + \mathbf{k}^\top \mathbf{P}\mathbf{k}}. \quad (128)$$

Let $\mathbf{P}_t^{\text{KF}}$ denote the covariance from the unprojected dense Kalman trajectory, and restrict the retained diagonal to the bounded positive family $\mathcal{P} = [p_{\min}, p_{\max}]^{d_k}$ with $0 < p_{\min} < p_{\max} < \infty$. Given a probe distribution ν_t over key-noise pairs (ℓ, r') applied immediately to that covariance, define

$$\mathbf{p}_t^{\text{GM}} \in \arg \min_{\mathbf{p} \in \mathcal{P}} \mathbb{E}_{(\ell, r') \sim \nu_t} \left[\Delta_t^{\text{KF}}(\ell, r') \left\| \mathbf{K}(\text{diag}(\mathbf{p}); \ell, r') - \mathbf{K}(\mathbf{P}_t^{\text{KF}}; \ell, r') \right\|_2^2 \right], \quad (129)$$

$$\Delta_t^{\text{KF}}(\ell, r') := r' + \ell^\top \mathbf{P}_t^{\text{KF}} \ell.$$

Unlike reverse or forward KL, Equation (129) asks for the diagonal state whose induced filtering action is closest to the exact action. A known transition and process-noise covariance can instead be incorporated by applying the full prediction map $\mathbf{P} \mapsto \mathbf{D}\mathbf{P}\mathbf{D}^\top + \mathbf{\Omega}$ to both covariances before evaluating the same objective. The probe law makes the required notion of closeness explicit: emphasizing the most recent key optimizes repeated-key gain fidelity to the dense filter, whereas a broader law protects gain fidelity across likely future observation-key directions. There is no distribution-free best diagonal state because different probes expose different missing cross-coordinate correlations.

The weight in Equation (129) follows from filtering risk rather than being chosen heuristically. For the exact dense predictive covariance $\hat{\mathbf{P}}_t^{\text{KF}}$, let

$$\Delta_t^{\text{KF}} = r_t + \mathbf{k}_t^\top \hat{\mathbf{P}}_t^{\text{KF}} \mathbf{k}_t, \quad \boldsymbol{\kappa}_t^{\text{KF}} = \mathbf{K}(\hat{\mathbf{P}}_t^{\text{KF}}; \mathbf{k}_t, r_t). \quad (130)$$

For one value column, applying an arbitrary gain $\mathbf{g}$ to the same predictive mean and prior has one-step mean-squared error

$$\mathcal{R}_t^{\text{KF}}(\mathbf{g}) = \text{Tr} \left[(\mathbf{I} - \mathbf{g}\mathbf{k}_t^\top) \hat{\mathbf{P}}_t^{\text{KF}} (\mathbf{I} - \mathbf{g}\mathbf{k}_t^\top)^\top + r_t \mathbf{g}\mathbf{g}^\top \right]. \quad (131)$$

Completing the square gives

$$\mathcal{R}_t^{\text{KF}}(\mathbf{g}) - \mathcal{R}_t^{\text{KF}}(\boldsymbol{\kappa}_t^{\text{KF}}) = \Delta_t^{\text{KF}} \|\mathbf{g} - \boldsymbol{\kappa}_t^{\text{KF}}\|_2^2. \quad (132)$$

Thus innovation-weighted gain error is not merely a convenient proxy: for this one-step problem, it is exactly the excess Bayes risk of the approximate mean update for every $\mathbf{g}$. Equivalently, since the innovation has variance Δ_t^{KF} , it is the expected squared distance between the approximate and exact posterior means. For d_v conditionally independent value columns, both sides are multiplied by d_v .

This identity suggests the direct constrained problem

$$\boldsymbol{\kappa}_t^\dagger \in \arg \min_{\mathbf{g} \in \bar{\mathcal{G}}_{\text{scan},t}} \Delta_t^{\text{KF}} \|\mathbf{g} - \boldsymbol{\kappa}_t^{\text{KF}}\|_2^2, \quad (133)$$

where $\bar{\mathcal{G}}_{\text{scan},t}$ is the closure of the gains realizable by the chosen fixed-state associative recurrence. The pointwise diagonal family induced by $\mathcal{P}$ is

$$\mathcal{G}_{\text{diag}}(\mathbf{k}_t, r_t; \mathcal{P}) = \{\mathbf{K}(\text{diag}(\mathbf{p}); \mathbf{k}_t, r_t) : \mathbf{p} \in \mathcal{P}\}. \quad (134)$$

Ignoring the finite bounds, its closure is

$$\bar{\mathcal{G}}_{\text{diag}}(\mathbf{k}_t) = \left\{ \mathbf{g} : g_i = \gamma_i k_{t,i}, \gamma_i \geq 0, \mathbf{k}_t^\top \mathbf{g} \leq 1 \right\}. \quad (135)$$

For finite strictly positive diagonal variances, $\gamma_i > 0$ on the support of $\mathbf{k}_t$ and $\mathbf{k}_t^\top \mathbf{g} < 1$. Moreover, the pointwise family in Equation (134) is only a relaxation of the gains jointly reachable by a shared recurrent scan state. The exact dense product $\hat{\mathbf{P}}_t^{\text{KF}} \mathbf{k}_t$ can rotate the key, reverse an individual coordinate sign, or be nonzero where $k_{t,i} = 0$. If the resulting gain violates the alignment conditions in Equation (135), no diagonal covariance can reproduce it exactly. Failure to lie only in the bounded family of Equation (134) instead means that no diagonal within the chosen numerical range can reproduce it. Ignoring those finite bounds, when a pointwise match exists its action-equivalent predictive diagonal,

$$\hat{p}_{t,i}^{\text{act}} = \frac{r_t \kappa_{t,i}^{\text{KF}}}{k_{t,i}(1 - \mathbf{k}_t^\top \boldsymbol{\kappa}_t^{\text{KF}})} = \frac{(\hat{\mathbf{P}}_t^{\text{KF}} \mathbf{k}_t)_i}{k_{t,i}}, \quad (136)$$

for coordinates with $k_{t,i} \neq 0$, is key dependent, may fail positivity, and is not a reusable approximation to the covariance. Computing it also already requires the dense covariance–key product that the approximation is intended to avoid.

There is a further causal distinction between optimizing a gain and optimizing the posterior state. The projected $\mathbf{p}_t$ is formed after $\boldsymbol{\kappa}_t$ has updated the memory, so changing $\mathbf{p}_t$ cannot improve the gain at the same token. It first changes the prediction and gain at token $t + 1$. For a candidate diagonal state $\mathbf{p}$, define

$$\begin{aligned}\widehat{\mathbf{P}}_{t+1}(\mathbf{p}) &= \mathbf{D}_{t+1} \text{diag}(\mathbf{p}) \mathbf{D}_{t+1}^\top + \boldsymbol{\Omega}_{t+1}, \\ \widehat{\mathbf{P}}_{t+1}^{\text{KF}} &= \mathbf{D}_{t+1} \mathbf{P}_t^{\text{KF}} \mathbf{D}_{t+1}^\top + \boldsymbol{\Omega}_{t+1}, \\ \boldsymbol{\kappa}_{t+1}(\mathbf{p}) &= \mathbf{K}(\widehat{\mathbf{P}}_{t+1}(\mathbf{p}); \mathbf{k}_{t+1}, r_{t+1}), \\ \Delta_{t+1}^{\text{KF}} &= r_{t+1} + \mathbf{k}_{t+1}^\top \widehat{\mathbf{P}}_{t+1}^{\text{KF}} \mathbf{k}_{t+1}, \quad \boldsymbol{\kappa}_{t+1}^{\text{KF}} = \mathbf{K}(\widehat{\mathbf{P}}_{t+1}^{\text{KF}}; \mathbf{k}_{t+1}, r_{t+1}).\end{aligned}\tag{137}$$

The gain-aware analogue of a one-step covariance projection is therefore

$$\mathbf{p}_t^\dagger \in \arg \min_{\mathbf{p} \in \mathcal{P}} \mathbb{E}_{\nu_t} [\Delta_{t+1}^{\text{KF}} \|\boldsymbol{\kappa}_{t+1}(\mathbf{p}) - \boldsymbol{\kappa}_{t+1}^{\text{KF}}\|_2^2],\tag{138}$$

where $\nu_t(\cdot \mid \mathcal{F}_t)$ is the conditional distribution of the next transition, process-noise covariance, key, and observation noise. A discounted multi-step version replaces the integrand by the corresponding sum over future gains. Equation (138) makes explicit that there is no distribution-free best diagonal covariance: the useful approximation depends on which future key directions will query the state.

A tractable local approximation. The exact objective is nonlinear because the candidate covariance appears in both the numerator and denominator of $\mathbf{K}$, and evaluating it against $\mathbf{P}_t^{\text{KF}}$ requires dense matrix–vector products. A tractable local surrogate replaces the globally rolled covariance by the exact one-step posterior $\mathbf{P}_t^*$ formed from the retained diagonal predictive prior. This target has the diagonal-minus-rank-one form in Equation (117). Suppressing the token index, for a probe $(\boldsymbol{\ell}, r')$, abbreviate

$$\boldsymbol{\kappa}^*(\boldsymbol{\ell}, r') = \mathbf{K}(\mathbf{P}^*; \boldsymbol{\ell}, r'), \quad \Delta^*(\boldsymbol{\ell}, r') = r' + \boldsymbol{\ell}^\top \mathbf{P}^* \boldsymbol{\ell}.\tag{139}$$

Linearizing the gain around $\mathbf{P}^*$ gives

$$\mathbf{K}(\text{diag}(\mathbf{p}); \boldsymbol{\ell}, r') - \boldsymbol{\kappa}^*(\boldsymbol{\ell}, r') \simeq \frac{(\mathbf{I} - \boldsymbol{\kappa}^*(\boldsymbol{\ell}, r') \boldsymbol{\ell}^\top)(\text{diag}(\mathbf{p}) - \mathbf{P}^*) \boldsymbol{\ell}}{\Delta^*(\boldsymbol{\ell}, r')}.\tag{140}$$

Define

$$\mathbf{B}(\boldsymbol{\ell}, r') = (\mathbf{I} - \boldsymbol{\kappa}^*(\boldsymbol{\ell}, r') \boldsymbol{\ell}^\top) \text{diag}(\boldsymbol{\ell}), \quad \mathbf{y}(\boldsymbol{\ell}, r') = (\mathbf{I} - \boldsymbol{\kappa}^*(\boldsymbol{\ell}, r') \boldsymbol{\ell}^\top) \mathbf{P}^* \boldsymbol{\ell}.\tag{141}$$

Substitution into Equation (129), with $\mathbf{P}_t^{\text{KF}}$ replaced by this local target, yields the nonnegative Jacobian-linearized quadratic problem

$$\mathbf{p}_t^{\text{lin}} \in \arg \min_{\mathbf{p} \in \mathcal{P}} \mathbb{E}_{\nu_t} \left[\frac{\|\mathbf{B}(\boldsymbol{\ell}, r') \mathbf{p} - \mathbf{y}(\boldsymbol{\ell}, r')\|_2^2}{\Delta^*(\boldsymbol{\ell}, r')} \right].\tag{142}$$

For a finite set of probe keys, this is a box-constrained least-squares problem. It retains the denominator sensitivity of the Kalman gain. Its normal matrix is generally dense, although for M probes it is a diagonal matrix plus a correction of rank at most $2M$.

A cheaper diagonal-Hessian approximation freezes the candidate denominator at $\Delta^*(\ell, r')$, equivalently dropping the left factor $\mathbf{I} - \kappa^* \ell^\top$ in Equation (140). The coordinates then decouple:

$$p_{t,i}^{\text{FD}} = \Pi_{[p_{\min}, p_{\max}]} \left(\frac{\mathbb{E}_{\nu_t}[\ell_i(\mathbf{P}^* \ell)_i / \Delta^*(\ell, r')]}{\mathbb{E}_{\nu_t}[\ell_i^2 / \Delta^*(\ell, r')]} \right). \quad (143)$$

Here $\Pi_{[p_{\min}, p_{\max}]}$ clamps the unconstrained coordinatewise minimizer to the retained variance range. If a coordinate has zero probe energy, the objective does not identify that coordinate and its previous or moment-matched value can be retained. When $\mathbf{P}_t^*$ is the exact one-step posterior in Equation (117), rename its incoming diagonal $\mathbf{p}^-$ and set $z = \sum_i p_i^- k_i^2$. Every required product is then available in linear time from

$$\mathbf{P}_t^* \ell = \mathbf{p}^- \odot \ell - \frac{(\mathbf{p}^- \odot \mathbf{k})(\mathbf{p}^- \odot \mathbf{k})^\top \ell}{r + z}. \quad (144)$$

Thus, for this local rank-one target, M probe directions cost $O(Md_k)$ without storing the dense covariance. A general exact-KLA trajectory instead requires dense matrix–vector products unless additional covariance structure is retained. For isotropic probes in the high-noise regime, where the weight $1/\Delta^*$ is approximately direction independent, Equation (143) reduces to the moment projection $p_{t,i} = P_{t,ii}^*$ when the box constraints are inactive; otherwise it gives the clipped moment solution. A point mass on the most recent key similarly gives the action-equivalent solution below when that solution is admissible. A mixture of these probe laws continuously trades global gain fidelity against repeated-key gain fidelity.

A closed-form repeated-key example. The distinction from scalar directional matching is already visible in the one-step setting of Equation (117). With no intervening dynamics or process noise, define

$$p_i^{\text{act}} = \frac{r}{r + z} p_i^-. \quad (145)$$

Then

$$\text{diag}(\mathbf{p}^{\text{act}}) \mathbf{k} = \mathbf{P}^* \mathbf{k} = \frac{r}{r + z} (\mathbf{p}^- \odot \mathbf{k}), \quad \mathbf{k}^\top \text{diag}(\mathbf{p}^{\text{act}}) \mathbf{k} = \mathbf{k}^\top \mathbf{P}^* \mathbf{k}. \quad (146)$$

When $\mathbf{p}^{\text{act}} \in \mathcal{P}$, it therefore reproduces the entire exact gain vector for the next identical key, for any next observation noise. Outside the chosen box it remains the unconstrained action-equivalent reference. In contrast, Equation (122) enforces only the second equality and thus matches only β^{eff} . The action-equivalent state is still not an exact covariance projection: on coordinates where $k_i \neq 0$ it is the unique diagonal action match, while coordinates with $k_i = 0$ are unconstrained. The uniform extension in Equation (145) chooses to shrink those unobserved coordinates as well. It has no guarantee for a different future key and couples all channels through z . It consequently cannot be inserted into the current independent Möbius maps without losing their input-only form.

Computational consequence. Equations (142) and (143) are local approximations to the ideal gain objective, not variational projections. For the one-step rank-one target they require only the structured products in Equation (144); a general dense target requires dense products. Their coefficients depend on the incoming recurrent state through $\mathbf{p}^-$, z , and Δ^* . The resulting update couples channels and therefore is not an input-only collection of scalar Möbius maps. The diagonal-Hessian rule reduces arithmetic and storage to $O(Md_k)$ for fixed M , but it does not by itself preserve the present associative gain scan.

To retain that scan, one can restrict the candidate in Equation (129) to the existing scaled family $\mathbf{p}(s)$ from Equation (121). A fixed or headwise s , or a prescribed token-input-only rule for s_t , preserves an input-only local map while matching only the part of the dense gain reachable by that one-dimensional family. The repeated-key approximation $s \simeq d_{\text{eff}}$ in Equation (123) is a useful initialization for matching the scalar effective gain; it solves the full vector objective only in balanced symmetric cases. Solving the exact gain objective for s_t from the recurrent covariance, or allowing the probe weights to depend on that state, gives a closer local solution but destroys the independent scan.

Finally, κ is invariant under the common scale gauge

$$\mathbf{P} \mapsto \lambda \mathbf{P}, \quad \mathbf{p} \mapsto \lambda \mathbf{p}, \quad r \mapsto \lambda r, \quad (p_{\min}, p_{\max}) \mapsto \lambda(p_{\min}, p_{\max}). \quad (147)$$

The risk weight Δ^* scales by λ . Under this simultaneous transformation the optimizer scales with λ and the objective is multiplied by a common constant. Holding the candidate box fixed would break this invariance. A fixed gauge, for example $r \equiv 1$, or normalized weights are useful when objectives with independently free scales are aggregated across heads or steps; unweighted gain error is gauge invariant but loses the exact excess-risk interpretation. Equation (132) is also a local identity for a common predictive mean and prior. Across multiple recursively projected steps, the gain objective remains a decision-aware approximation criterion, but it is not equal to the total trajectory-level filtering error.

Gain matching seeks fidelity to the exact Kalman action; it does not deliberately make that action more conservative. If the task prefers additional retention beyond the approximate Kalman operating point, that preference should be introduced separately rather than hidden inside a covariance projection.

8.3 Learning confidence as a bounded protection state

The gain-matching objective addresses approximation error, whereas the observed benefit of a larger information scale points to a different failure mode: the carried state does not protect past writes strongly enough from later updates. The appropriate intervention should therefore increase the confidence passed forward, rather than merely truncate the current gain after it has been computed. A larger protection scale changes this retention–plasticity operating point, whereas the $B = 2$ channelwise cap changes only those current gains for which $\beta_{t,i}^{\text{ch}} > 2$. The cap can therefore be inactive even when repeated writes are too strong; when it is active, it can remove useful anisotropy without materially changing $\beta_t^{\text{eff}} = \sum_i k_{t,i}^2 \beta_{t,i}^{\text{ch}}$. Its failure to improve accuracy is thus evidence against extreme channelwise amplification as the primary bottleneck, not against the broader overwrite hypothesis. For the same reason, an off-key projection cannot reproduce the benefit of a larger information scale: by construction it preserves β_t^{eff} . This diagnosis should be accompanied by the cap activation rate and the removed effective-write mass; a nearly inactive cap is a null intervention rather than evidence about the mechanism.

A more principled scan-compatible response is to make the retention–plasticity tradeoff explicit in the inference objective rather than clipping the resulting gain. Let $\rho_h \geq 1$ denote a protection-to-write ratio for head h and recall $V_t = \mathbf{k}_t^\top \text{diag}(\mathbf{p}_t) \mathbf{k}_t$. Starting from the ordinary reverse-KL objective $\overline{\mathcal{J}}_{t,j}^{(1)}$ in Equation (61), define

$$\mathcal{J}_{t,j}^{\text{ret}}(q_{t,j}; \rho_h) = \overline{\mathcal{J}}_{t,j}^{(1)}(q_{t,j}) + \frac{\rho_h - 1}{2r_t} V_t. \quad (148)$$

The additional term prices uncertainty remaining in the direction that has just been written. Its unique minimizer leaves the current posterior-mean correction unchanged and strengthens only the

precision retained for future tokens:

$$\hat{c}_{t,i} = \frac{c_{t-1,i}}{\alpha_{t,i}^2 + \omega_{t,i}c_{t-1,i}}, \quad \hat{p}_{t,i} = \hat{c}_{t,i}^{-1}, \quad \kappa_t = \frac{\hat{\mathbf{p}}_t \odot \mathbf{k}_t}{r_t + \sum_i \hat{p}_{t,i} k_{t,i}^2}, \quad \mathbf{c}_t = \hat{\mathbf{c}}_t + \rho_h \frac{\mathbf{k}_t \odot \mathbf{k}_t}{r_t}. \quad (149)$$

Consequently, ρ_h does not alter the write at the token that supplies the evidence, but it reduces the gains of later tokens whose coordinatewise key energy overlaps that evidence. This is consistent with the causal effect indicated when a larger information scale improves accuracy. It is also the $(s_{\text{write}}, s_{\text{protect}}) = (1, \rho_h)$ specialization of the split variational objective in Equation (55).

Because protection is a multiplicative scale, the smallest useful parameterization is one bounded log-scale scalar per head,

$$\rho_h = \exp\left(\log \rho_{\min} + \sigma(\theta_h) \log \frac{\rho_{\max}}{\rho_{\min}}\right), \quad 1 \leq \rho_{\min} < \rho_{\max}, \quad (150)$$

initialized at the best fixed-scale operating point. A token-dependent $\rho_{t,h}$ remains scan-compatible provided that it depends only on the current token, but the headwise form is more identifiable and is the cleaner first experiment. Equation (149) only changes the token-local coefficients of the Möbius maps, so both the uncertainty scan and the affine memory scan are preserved. A positive process-noise floor remains necessary to restore plasticity and prevent the learned protection ratio from freezing the memory.

To encourage rather than merely permit confidence, initialize ρ_h at a fixed value $\rho_0 > 1$ selected by the information-scale sweep and optionally use the weak log-scale anchor

$$\mathcal{L}_\rho = \lambda_\rho \sum_h (\log \rho_h - \log \rho_0)^2. \quad (151)$$

Regularizing the raw state $\mathbf{c}_t$ itself is less suitable: its magnitude grows with the evidence history and participates in a common covariance–noise scale symmetry. The global scale should first be fixed, for example with $r_t \equiv 1$ and a fixed prior scale, so that ρ_h has an identifiable retention meaning. A bounded ρ_h , together with $\omega_{t,i} \geq \omega_{\min} > 0$, lets training choose stronger protection without reaching the degenerate zero-plasticity limit. Reducing ω_t would also increase confidence, but it changes the assumed drift of the latent memory and is therefore a less targeted intervention.

For $\rho_h = 1$, Equation (148) reduces to ordinary variational filtering. For $\rho_h > 1$, it is a decision-aware variational objective rather than the ELBO of the original linear–Gaussian observation model: a single Gaussian likelihood cannot leave the current mean update unchanged while increasing only future confidence. If exact Bayesian semantics are required, the write and protection scales must instead be tied and the process and observation noise calibrated under that coherent model. Under the fixed-size associative-memory objective, however, ρ_h has a direct interpretation as the cost of future overwrite. The retained precision should then be described as a gain-conditioning or protection state, while channelwise and off-key caps remain optional worst-case guardrails rather than the primary retention mechanism.

Separating a reference uncertainty state from protection. If an independently testable uncertainty approximation and overwrite control are both useful, a more explicit design carries two conceptually distinct states. Let $\mathbf{P}_t^{\text{unc}}$ be a fidelity-oriented reference covariance obtained from either the moment projection or the scan-compatible process-noise-floor rule. Given its prediction, define

$$\kappa_t^{\text{unc}} = \frac{\hat{\mathbf{P}}_t^{\text{unc}} \mathbf{k}_t}{\Delta_t^{\text{unc}}}, \quad \Delta_t^{\text{unc}} = r_t + \mathbf{k}_t^\top \hat{\mathbf{P}}_t^{\text{unc}} \mathbf{k}_t. \quad (152)$$

For one value column and an arbitrary linear gain $\mathbf{g}$, use the one-step reference risk

$$\begin{aligned}\mathcal{R}_t^{\text{ref}}(\mathbf{g}) &= \text{Tr} \left[(\mathbf{I} - \mathbf{g}\mathbf{k}_t^\top) \hat{\mathbf{P}}_t^{\text{unc}} (\mathbf{I} - \mathbf{g}\mathbf{k}_t^\top)^\top + r_t \mathbf{g}\mathbf{g}^\top \right] \\ &= \mathcal{R}_t^{\text{ref}}(\boldsymbol{\kappa}_t^{\text{unc}}) + \Delta_t^{\text{unc}} \|\mathbf{g} - \boldsymbol{\kappa}_t^{\text{unc}}\|_2^2.\end{aligned}\tag{153}$$

The second state records the importance of past key directions. The full discounted Gram state would be

$$\hat{\mathbf{A}}_t = \eta_t \mathbf{A}_{t-1}, \quad \mathbf{A}_t = \hat{\mathbf{A}}_t + \frac{\xi_t}{r_t} \mathbf{k}_t \mathbf{k}_t^\top, \quad 0 < \eta_t \leq 1, \quad \xi_t \geq 0.\tag{154}$$

For the rank-one memory change $\Delta \mathbf{S}_t = \mathbf{g} \mathbf{e}_t^\top$, its discounted squared effect at the protected keys is proportional to $\|\mathbf{e}_t\|_2^2 \mathbf{g}^\top \hat{\mathbf{A}}_t \mathbf{g}$. Absorbing an expected or normalized residual scale into λ_{ow} , the regularized decision rule is

$$\begin{aligned}\bar{\boldsymbol{\kappa}}_t &= \arg \min_{\mathbf{g}} \left\{ \mathcal{R}_t^{\text{ref}}(\mathbf{g}) + \lambda_{\text{ow}} \mathbf{g}^\top \hat{\mathbf{A}}_t \mathbf{g} \right\} \\ &= \left(\Delta_t^{\text{unc}} \mathbf{I} + \lambda_{\text{ow}} \hat{\mathbf{A}}_t \right)^{-1} \Delta_t^{\text{unc}} \boldsymbol{\kappa}_t^{\text{unc}}.\end{aligned}\tag{155}$$

This penalizes a write according to its predicted interference with previously important keys rather than according to an instantaneous channel threshold.

The full Gram state costs $O(d_k^2)$. Its diagonal approximation,

$$\hat{\mathbf{a}}_t = \eta_t \mathbf{a}_{t-1}, \quad \mathbf{a}_t = \hat{\mathbf{a}}_t + \xi_t \frac{\mathbf{k}_t \odot \mathbf{k}_t}{r_t}, \quad \bar{\kappa}_{t,i} = \frac{\Delta_t^{\text{unc}}}{\Delta_t^{\text{unc}} + \lambda_{\text{ow}} \hat{a}_{t,i}} \kappa_{t,i}^{\text{unc}},\tag{156}$$

is a basis-dependent surrogate for key-overlap interference, but its recurrence is an associative affine scan whenever η_t and ξ_t depend only on the current token. An isotropic $\hat{\mathbf{A}}_t = a_t \mathbf{I}$ instead gives basis-invariant radial shrinkage. Combined with the process-noise-floor approximation, the diagonal design requires one Möbius uncertainty scan and one affine importance scan before the unchanged affine memory scan, and remains $O(d_k)$ in recurrent state. A moment-projected uncertainty state does not retain this scan property.

There is an unavoidable consistency caveat. Once $\bar{\boldsymbol{\kappa}}_t \neq \boldsymbol{\kappa}_t^{\text{unc}}$, $\mathbf{P}_t^{\text{unc}}$ is a counterfactual reference state, not the error covariance around the actual regularized mean. The covariance consistent with that mean would use the Joseph update

$$\mathbf{P}_t^{\text{err}} = (\mathbf{I} - \bar{\boldsymbol{\kappa}}_t \mathbf{k}_t^\top) \hat{\mathbf{P}}_t^{\text{err}} (\mathbf{I} - \bar{\boldsymbol{\kappa}}_t \mathbf{k}_t^\top)^\top + r_t \bar{\boldsymbol{\kappa}}_t \bar{\boldsymbol{\kappa}}_t^\top.\tag{157}$$

Projecting this covariance feeds the protection state back into the uncertainty recurrence and generally loses the independent scalar Möbius scan. The two-state construction is therefore an honest decision-aware controller with a reference uncertainty model, not an exact Bayesian filter. If covariance around the actual mean must remain calibrated, the structured or sequential alternatives above are required. Otherwise the single bounded ρ_h protection state is the lower-cost design supported most directly by the present ablation.

8.4 A bounded residual can hide an unbounded channel gain

Let

$$\Delta_t = r_t + \sum_j \hat{p}_{t,j} k_{t,j}^2, \quad \beta_{t,i}^{\text{ch}} = \frac{\hat{p}_{t,i}}{\Delta_t}, \quad \kappa_{t,i} = \beta_{t,i}^{\text{ch}} k_{t,i}.\tag{158}$$

Here $\beta_{t,i}^{\text{ch}}$ is the per-channel write multiplier, not a probability or a residual-contraction fraction. For an ℓ_2 -normalized key, its key-energy-weighted average is the effective write strength

$$\beta_t^{\text{eff}} = \mathbf{k}_t^\top \boldsymbol{\kappa}_t = \sum_i \beta_{t,i}^{\text{ch}} k_{t,i}^2 = \frac{\sum_i \hat{p}_{t,i} k_{t,i}^2}{\Delta_t} \in [0, 1). \quad (159)$$

Consequently the residual at the current key contracts safely by $1 - \beta_t^{\text{eff}}$. This scalar guarantee does not bound any individual $\beta_{t,i}^{\text{ch}}$, the gain norm, or the effect at another key.

The failure is structural. Fix $b \in (0, 1)$, set $z = b/(1 - b)$ and $r_t = 1$, and consider

$$\mathbf{k}_\epsilon = [\epsilon \quad \sqrt{1 - \epsilon^2}]^\top, \quad \hat{\mathbf{p}}_\epsilon = \begin{bmatrix} z & z \\ 2\epsilon^2 & 2(1 - \epsilon^2) \end{bmatrix}^\top. \quad (160)$$

For every $\epsilon \in (0, 1)$, the key is normalized and $\sum_i \hat{p}_i k_i^2 = z$, so $\beta^{\text{eff}} = b$ is fixed. Nevertheless,

$$\beta_1^{\text{ch}} = \frac{b}{2\epsilon^2} \longrightarrow \infty, \quad \kappa_1 = \frac{b}{2\epsilon} \longrightarrow \infty \quad (\epsilon \rightarrow 0). \quad (161)$$

A small key coordinate can therefore hide arbitrarily large channel uncertainty from the shared denominator. With residual $\mathbf{e}_t = \mathbf{v}_t - \tilde{\mathbf{S}}_t^\top \mathbf{k}_t$, the induced change at a unit read key $\mathbf{q}$ is

$$(\mathbf{S}_t - \tilde{\mathbf{S}}_t)^\top \mathbf{q} = \mathbf{e}_t (\boldsymbol{\kappa}_t^\top \mathbf{q}). \quad (162)$$

Thus the same-key residual can remain contractive while an off-key read, or the nonnormal erase operator $\mathbf{I} - \boldsymbol{\kappa}_t \mathbf{k}_t^\top$, is arbitrarily amplified. This is the channelwise overwrite failure that β_t^{eff} alone cannot detect.

8.5 Capping the channelwise write strength

A direct trust region clips the channel multipliers before the memory update:

$$\bar{\beta}_{t,i}^{\text{ch}} = \text{clip}_{[0,B]}(\beta_{t,i}^{\text{ch}}), \quad \bar{\kappa}_{t,i} = \bar{\beta}_{t,i}^{\text{ch}} k_{t,i}. \quad (163)$$

The lower bound is automatic under the positive covariance model; it is written explicitly to state the admissible gain class. Because the cap only decreases nonnegative coefficients,

$$0 \leq \bar{\beta}_t^{\text{eff}} := \mathbf{k}_t^\top \bar{\boldsymbol{\kappa}}_t \leq \beta_t^{\text{eff}} < 1. \quad (164)$$

The capped update therefore preserves same-key residual contraction and cannot overshoot that residual. More importantly, unit-normalized keys give the new gain-norm bound

$$\|\bar{\boldsymbol{\kappa}}_t\|_2^2 = \sum_i (\bar{\beta}_{t,i}^{\text{ch}})^2 k_{t,i}^2 \leq B \sum_i \bar{\beta}_{t,i}^{\text{ch}} k_{t,i}^2 = B \bar{\beta}_t^{\text{eff}} < B. \quad (165)$$

For the proposed $B = 2$, the gain norm is strictly below $\sqrt{2}$. If $\bar{\boldsymbol{\kappa}}_t = \bar{\beta}_t^{\text{eff}} \mathbf{k}_t + \bar{\boldsymbol{\kappa}}_{t,\perp}$, then

$$\|\bar{\boldsymbol{\kappa}}_{t,\perp}\|_2^2 \leq 2\bar{\beta}_t^{\text{eff}} - (\bar{\beta}_t^{\text{eff}})^2 \leq 1. \quad (166)$$

Moreover, $\|(\mathbf{I} - \bar{\boldsymbol{\kappa}}_t \mathbf{k}_t^\top) \mathbf{k}_t\|_2 \leq 1$ and $\|\mathbf{I} - \bar{\boldsymbol{\kappa}}_t \mathbf{k}_t^\top\|_2 \leq \sqrt{2}$. The operator need not be nonexpansive when an off-key component remains, but it is now uniformly bounded. The realized correction at any unit read key satisfies

$$\|(\mathbf{S}_t - \tilde{\mathbf{S}}_t)^\top \mathbf{q}\|_2 \leq \sqrt{2} \|\mathbf{e}_t\|_2. \quad (167)$$

The value $B = 2$ is also permissive around the isotropic regime: if $\hat{p}_{t,i} = \hat{p}_t$ for every channel, then $\beta_{t,i}^{\text{ch}} = \hat{p}_t / (r_t + \hat{p}_t) < 1$, so the cap is inactive. It targets only anisotropic amplification beyond the scalar-gain range.

The cap has a constrained estimation interpretation. For any linear gain $\mathbf{g}$, let $\mathcal{R}_t(\mathbf{g})$ be the trace of its conditional estimation-error covariance. Completing the square around the Kalman gain $\boldsymbol{\kappa}_t$ gives

$$\mathcal{R}_t(\mathbf{g}) = \mathcal{R}_t(\boldsymbol{\kappa}_t) + \Delta_t \|\mathbf{g} - \boldsymbol{\kappa}_t\|_2^2. \quad (168)$$

Minimizing this risk over $g_i = \gamma_i k_{t,i}$ with $0 \leq \gamma_i \leq B$ yields $\gamma_i = \text{clip}_{[0,B]}(\beta_{t,i}^{\text{ch}})$, exactly Equation (163). The cap is therefore the minimum-risk linear correction within this bounded channel-aligned class.

This constrained correction is no longer the unconstrained Kalman posterior mean when the cap is active. Retaining the original information update also leaves the confidence state unchanged even though less of the residual was assimilated, so the mean and covariance no longer describe one Gaussian posterior. The cap should therefore be presented as an overwrite guardrail, not as a solution to reverse-KL overconfidence. A covariance update consistent with the capped gain would use its Joseph error covariance and would generally couple channels, forfeiting the independent Möbius scan. The gain-only cap is token-local after that scan and before the affine memory scan, so it preserves the current parallel algorithm. A hard cap has zero gain-path gradient in saturated channels; a smooth saturation is an alternative if this impedes training.

8.6 Gain-norm and off-key control

The channelwise cap in Equation (163) acts in the coordinate system of the diagonal covariance. Two complementary alternatives operate directly on the gain vector: a radial cap bounds the entire update, while an off-key cap changes only the component responsible for anisotropic cross-key interference.

A basis-invariant radial cap projects the gain onto an ℓ_2 ball. Define

$$m_t = \text{RMS}(\boldsymbol{\kappa}_t) = \frac{\|\boldsymbol{\kappa}_t\|_2}{\sqrt{d_k}}, \quad a_t^{\text{rad}} = \min\left\{1, \frac{\tau_{\text{rms}}}{m_t}\right\}, \quad \boldsymbol{\kappa}_t^{\text{rad}} = a_t^{\text{rad}} \boldsymbol{\kappa}_t, \quad (169)$$

with $a_t^{\text{rad}} = 1$ when $m_t = 0$. This one-sided RMS cap is exactly radial norm clipping with radius $\tau_2 = \sqrt{d_k} \tau_{\text{rms}}$. It preserves the gain direction and satisfies

$$\|\boldsymbol{\kappa}_t^{\text{rad}}\|_2 \leq \tau_2, \quad \beta_{t,\text{rad}}^{\text{eff}} = a_t^{\text{rad}} \beta_t^{\text{eff}} \leq \beta_t^{\text{eff}} < 1. \quad (170)$$

It therefore bounds every cross-key correction, but also weakens the intended same-key write whenever active. Choosing $\tau_{\text{rms}} = \sqrt{2/d_k}$ matches the $\|\bar{\boldsymbol{\kappa}}_t\|_2 < \sqrt{2}$ ceiling of the $B = 2$ channel cap. A raw RMS threshold of 2 would instead permit norm $2\sqrt{d_k}$ and is not comparable to a channel cap of 2.

To change only the anisotropic component, decompose

$$\boldsymbol{\kappa}_{t,\perp} = \boldsymbol{\kappa}_t - \beta_t^{\text{eff}} \mathbf{k}_t, \quad a_t^\perp = \min\left\{1, \frac{\tau_{\perp,t}}{\|\boldsymbol{\kappa}_{t,\perp}\|_2}\right\}, \quad \boldsymbol{\kappa}_t^{\text{off}} = \beta_t^{\text{eff}} \mathbf{k}_t + a_t^\perp \boldsymbol{\kappa}_{t,\perp}, \quad (171)$$

taking $a_t^\perp = 1$ when the off-key norm is zero. This preserves $\mathbf{k}_t^\top \boldsymbol{\kappa}_t^{\text{off}} = \beta_t^{\text{eff}}$ exactly, so the current residual contraction is unchanged, while every unit $\mathbf{q} \perp \mathbf{k}_t$ obeys $|\mathbf{q}^\top \boldsymbol{\kappa}_t^{\text{off}}| \leq \tau_{\perp,t}$. A fixed $\tau_\perp$ gives a direct cross-talk budget. The adaptive choice

$$\tau_{\perp,t} = \sqrt{2\beta_t^{\text{eff}} - (\beta_t^{\text{eff}})^2} \quad (172)$$

yields $\|\kappa_t^{\text{off}}\|_2^2 \leq 2\beta_t^{\text{eff}}$ and $\|(\mathbf{I} - \kappa_t^{\text{off}} \mathbf{k}_t^\top) \mathbf{k}_t\|_2 \leq 1$, matching the useful $B = 2$ envelope without changing the scalar write strength. Setting $\tau_{\perp,t} = 0$ recovers the key-aligned gain $\beta_t^{\text{eff}} \mathbf{k}_t$; a learned shrinkage between these extremes is the continuous version of the $\eta \in \{0, 1\}$ ablation in Equation (110).

Full RMS normalization is qualitatively different and is not recommended here. For a nonzero gain, setting

$$\kappa_t^{\text{RMSNorm}} = \frac{\rho}{\text{RMS}(\kappa_t)} \kappa_t \quad (173)$$

forces $\|\kappa_t^{\text{RMSNorm}}\|_2 = \rho\sqrt{d_k}$ and replaces the effective write by $\rho\sqrt{d_k} \beta_t^{\text{eff}} / \|\kappa_t\|_2$. It rescales small and large gains alike and can amplify a nearly zero gain that the uncertainty state intentionally suppressed. In the isotropic case $\kappa_t = \beta_t^{\text{eff}} \mathbf{k}_t$, every nonzero write becomes $\beta_{t,\text{RMSNorm}}^{\text{eff}} = \rho\sqrt{d_k}$, independent of its original confidence. Hence a conventional unit-RMS normalization can destroy residual contraction. The one-sided cap in Equation (169) is preferable because it never amplifies a gain. Both norm controls remain token-local and preserve the two scans when the information recurrence is left unchanged; when active, however, they share the mean-covariance inconsistency noted above.

8.7 Available controls and what each one solves

Together with the channelwise and norm controls above, the remaining principal interventions are as follows.

Future-protection scale. Increasing s_{protect} leaves the current write unchanged locally and suppresses later gains. It directly addresses repeated overwrite, but deliberately increases confidence and can freeze rewriting if taken too far.

Coherent current-write temperature. Increasing r_t , or equivalently reducing a likelihood-precision multiplier in both the gain and posterior update, weakens current assimilation under a valid Gaussian observation model. Because it also leaves more posterior uncertainty, it need not protect against later writes.

Effective-write cap. Rescaling the complete gain by $\min\{1, \beta_{\text{max}}/\beta_t^{\text{eff}}\}$ for $0 < \beta_{\text{max}} < 1$ directly limits same-key overwrite while preserving the gain direction. It does not control a large off-key component when β_t^{eff} is already below the threshold.

Better calibrated uncertainty. Reducing s_{protect} , inflating the retained covariance, or keeping a positive process noise floor prevents the state from becoming too confident. The process-floor approximation is the tighter scan-compatible projection developed above; the moment projection matches the exact one-step marginal variances but loses the independent Möbius scan; structured or dense covariance tracking is more faithful still and more expensive. These changes can increase future gains, so better posterior calibration need not reduce overwrite.

Noise-scale and gauge controls. Fixing $r = 1$ and $\mu = 1$, bounding relative observation reliability, and starting process noise isotropically improve identifiability and avoid extreme gain temperatures. Reducing process noise can suppress future writes but risks covariance windup; increasing transition decay is not clean protection because it erases the stored memory directly.

Residual, update, or erase clipping. Clipping $\|e_t\|_2$ or the realized rank-one update controls large values as well as large gains. Separating the write and erase gates can limit destructive editing directly. Both options depart further from the Kalman residual update, and residual-dependent nonlinearities generally lose the input-only affine memory scan.

The clean experiment crosses one confidence-state intervention ($s = 1$, $s = d_k$, process-floor, or moment projection) with one current-write intervention (uncapped, an effective-write cap, the $B = 2$ channel cap, or off-key shrinkage). It should report β_t^{eff} , $\max_i \beta_{t,i}^{\text{ch}}$, $\|\kappa_t\|_2$, $\|\kappa_{t,\perp}\|_2$, cap activation rate, residual magnitude, and write-erase magnitude. This factorial comparison separates posterior calibration from overwrite control instead of asking one scalar information scale to solve both.

9 Mamba as a Controlled-Prediction Specialization

The zero-mean process model used for KDN does not, as written, contain the additive input write of Mamba. The minimal extension is to include a known token-dependent control in the state dynamics,

$$\mathbf{S}_t^* = \mathbf{D}_t \mathbf{S}_{t-1}^* + \mathbf{U}_t + \widetilde{\mathbf{W}}_t, \quad \widetilde{\mathbf{W}}_t \sim \mathcal{N}_{\text{col}}(\mathbf{0}, \mathbf{\Omega}_t). \quad (174)$$

Conditioned on the token-generated coefficients, both the transition and $\mathbf{U}_t$ are known, so the recurrence remains linear and $\mathbf{U}_t$ is deterministic. Equivalently, the control is the conditional mean of the original process increment, separated from its centered random part. Consequently, it changes the predicted memory but not its covariance:

$$\widehat{\mathbf{S}}_t = \mathbf{D}_t \mathbf{S}_{t-1} + \mathbf{U}_t, \quad \widehat{\mathbf{p}}_t = \alpha_t^2 \odot \mathbf{p}_{t-1} + \omega_t. \quad (175)$$

This is therefore a controlled linear-Gaussian state-space model with exactly the same uncertainty scan as KDN.

The per-head SSD/Mamba-2 core can be written in matrix-state form as

$$\mathbf{H}_t = a_t \mathbf{H}_{t-1} + \mathbf{B}_t \mathbf{x}_t^\top, \quad \mathbf{o}_t = \mathbf{H}_t^\top \mathbf{C}_t + \mathbf{D}_{\text{skip}} \mathbf{x}_t. \quad (176)$$

where the discretization scale is absorbed into $\mathbf{B}_t$. It is recovered from Equation (174) by setting $\mathbf{S}_t^* = \mathbf{H}_t$, $\mathbf{D}_t = a_t \mathbf{I}$, $\mathbf{U}_t = \mathbf{B}_t \mathbf{x}_t^\top$, and $\mathbf{\Omega}_t = \mathbf{0}$, with $\mathbf{C}_t$ serving as the read query. The direct path, output gates, and surrounding projections sit outside this recurrent state equation. Mamba-1 fits the same template after vectorizing its channel- and state-dependent diagonal dynamics. Mamba-3 SISO replaces the single rank-one control by a trapezoidal two-token control of the form $\mathbf{U}_t = \beta_t \mathbf{B}_{t-1} \mathbf{x}_{t-1}^\top + \gamma_t \mathbf{B}_t \mathbf{x}_t^\top$, while its MIMO variant uses a finite-rank sum of such controls. These remain deterministic controls once their token-generated coefficients are given.

This mapping is exact at the recurrent-state level, but it does not make Mamba a nontrivial special case of the Kalman posterior update. With the controlled prediction above, KDN would still apply

$$\mathbf{S}_t = \widehat{\mathbf{S}}_t + \kappa_t \left(\mathbf{v}_t - \widehat{\mathbf{S}}_t^\top \mathbf{k}_t \right)^\top. \quad (177)$$

The standard Mamba recurrence omits this observation-correction step: its additive write is a known control, its $\mathbf{C}_t$ is a read query rather than a measurement used to condition the state, and it carries no explicit filtering covariance under this embedding. Trying instead to reinterpret the Mamba write as an observation correction by setting $\kappa_t = \mathbf{B}_t$ and $\mathbf{v}_t = \mathbf{x}_t$ would introduce the additional erase term $-\kappa_t \mathbf{k}_t^\top \mathbf{D}_t \mathbf{S}_{t-1}$, which is absent from Mamba.

The precise unification is thus a division of roles within the controlled state-space template. Mamba is its zero-noise, prediction-only specialization, with $\mathbf{U}_t \neq \mathbf{0}$ and no observation correction. Delta-rule models set $\mathbf{U}_t = \mathbf{0}$ and prescribe a residual gain, whereas KDN derives that gain from propagated uncertainty. A combined model could use both a selective-SSM control and an uncertainty-weighted innovation, but that hybrid is a new model rather than an alternative interpretation of existing Mamba. Accordingly, the defensible paper-level claim is that Mamba and KDN share a controlled matrix-state scaffold and specialize complementary prediction and correction branches—not that Mamba itself performs Kalman filtering.

10 Predictive Innovation Likelihood and Covariance Calibration

The language-modeling objective can learn useful uncertainty-dependent gates without making their numerical values calibrated variances. A direct way to test, and optionally train, the probabilistic interpretation is to score the current value under the one-step predictive distribution. This criterion is distinct from the expected likelihood inside the variational objective above: it uses the uncertainty before the current observation is assimilated and evaluates the marginal distribution of that observation.

10.1 Marginal likelihood of the pre-write innovation

For value coordinate j , the predicted state and observation model are

$$\mathbf{s}_{t,j}^* \mid \mathcal{F}_{t-1} \sim \mathcal{N}(\hat{\mathbf{s}}_{t,j}, \hat{\mathbf{P}}_t), \quad v_{t,j} = \mathbf{k}_t^\top \mathbf{s}_{t,j}^* + e_{t,j}, \quad e_{t,j} \sim \mathcal{N}(0, r_t). \quad (178)$$

Marginalizing the uncertain state gives

$$v_{t,j} \mid \mathcal{F}_{t-1}, \mathbf{k}_t \sim \mathcal{N}(\hat{v}_{t,j}, \Delta_t), \quad \hat{v}_{t,j} = \mathbf{k}_t^\top \hat{\mathbf{s}}_{t,j}, \quad \Delta_t = r_t + \mathbf{k}_t^\top \hat{\mathbf{P}}_t \mathbf{k}_t. \quad (179)$$

Thus Δ_t combines irreducible observation noise with uncertainty in the predicted memory along the current key. Under the columnwise model, all d_v value coordinates share this scalar variance. With

$$\boldsymbol{\delta}_t = \mathbf{v}_t - \hat{\mathbf{S}}_t^\top \mathbf{k}_t, \quad (180)$$

their joint predictive negative log likelihood is

$$\mathcal{L}_{\text{pred},t} = \frac{\|\boldsymbol{\delta}_t\|_2^2}{2\Delta_t} + \frac{d_v}{2} \log \Delta_t + \frac{d_v}{2} \log(2\pi). \quad (181)$$

An equivalent dimension-normalized loss, up to an additive constant, is

$$\bar{\mathcal{L}}_{\text{pred},t} = \frac{1}{2} \left(\frac{d_v^{-1} \|\boldsymbol{\delta}_t\|_2^2}{\Delta_t} + \log \Delta_t \right). \quad (182)$$

Both the innovation and its variance must use the exclusive, pre-write state. Evaluating the residual after the update would let the current value help predict itself and would systematically understate the forecasting error.

The two NLL terms prevent opposite degeneracies. The standardized squared error $\|\boldsymbol{\delta}_t\|_2^2/\Delta_t$ penalizes a variance that is too small for the realized innovation, while $d_v \log \Delta_t$ penalizes variance inflation. In particular,

$$\frac{\partial \mathcal{L}_{\text{pred},t}}{\partial \log \Delta_t} = \frac{1}{2} \left(d_v - \frac{\|\boldsymbol{\delta}_t\|_2^2}{\Delta_t} \right), \quad \Delta_t^* = \frac{\|\boldsymbol{\delta}_t\|_2^2}{d_v} \quad \text{for a fixed realized innovation.} \quad (183)$$

Across repeated contexts, the population optimum is the conditional mean squared innovation per value coordinate. The log term is essential whenever an uncertainty scale is learned: omitting it leaves the objective free to reduce the standardized error by sending the variance to infinity.

10.2 Diagonal and isotropic specializations

For Diagonal KDN, the predictive covariance is $\hat{\mathbf{P}}_t = \text{diag}(\hat{\mathbf{p}}_t)$, so the scored innovation variance is

$$\Delta_t^{\text{diag}} = r_t + \sum_i \hat{p}_{t,i} k_{t,i}^2. \quad (184)$$

This is exactly the scalar denominator of the Kalman gain. The predictive likelihood therefore tests whether the same uncertainty that controls the write also predicts the observed residual scale. It scores only the directional covariance visible at the current key; it cannot by itself detect covariance error in directions that the observed keys do not probe.

For Isotropic KDN, $\hat{\mathbf{P}}_t = \hat{b}_t \mathbf{I}$ gives

$$\Delta_t^{\text{iso}} = r_t + \hat{b}_t \|\mathbf{k}_t\|_2^2, \quad (185)$$

which reduces to $r_t + \hat{b}_t$ for unit-normalized keys. The same NLL can therefore compare the two covariance families without changing their state budgets: each model is scored using the predictive variance implied by its own retained state.

10.3 Relation to the VAE objective and variational expected NLL

The connection to a standard variational autoencoder (VAE) objective is algebraic. For one value coordinate, let $\hat{\pi}_{t,j} = p(\mathbf{s}_{t,j}^* | \mathcal{F}_{t-1})$ denote the pre-write predictive prior, and let $q_{t,j} = \mathcal{N}(\boldsymbol{\mu}_{t,j}, \mathbf{P}_t)$ be a candidate filtering posterior that may depend on the current observation $v_{t,j}$. The negative ELBO has the familiar reconstruction-plus-regularization form

$$-\text{ELBO}_{t,j}(q_{t,j}) = \underbrace{\mathbb{E}_{q_{t,j}}[-\log p(v_{t,j} | \mathbf{s}_{t,j}^*, \mathbf{k}_t)]}_{\text{expected reconstruction NLL}} + \underbrace{\text{KL}(q_{t,j} \parallel \hat{\pi}_{t,j})}_{\text{regularization to the predictive prior}}. \quad (186)$$

For the Gaussian observation model, the reconstruction term is

$$\mathbb{E}_{q_{t,j}}[-\log p(v_{t,j} | \mathbf{s}_{t,j}^*, \mathbf{k}_t)] = \frac{1}{2r_t} \left[\left(v_{t,j} - \mathbf{k}_t^\top \boldsymbol{\mu}_{t,j} \right)^2 + \mathbf{k}_t^\top \mathbf{P}_t \mathbf{k}_t \right] + \frac{1}{2} \log(2\pi r_t). \quad (187)$$

This term is evaluated under a post-observation distribution: $q_{t,j}$ is selected after $v_{t,j}$ has been observed, so its mean may already incorporate the value being reconstructed. The KL term charges the posterior for moving away from the pre-write prior. The reconstruction term alone is therefore not a one-step forecasting score.

By contrast, the predictive innovation NLL marginalizes the latent state under the pre-write prior,

$$\mathcal{L}_{\text{pred},t,j} = -\log p(v_{t,j} | \mathcal{F}_{t-1}, \mathbf{k}_t) = -\log \int p(v_{t,j} | \mathbf{s}, \mathbf{k}_t) \hat{\pi}_{t,j}(\mathbf{s}) d\mathbf{s}. \quad (188)$$

It therefore evaluates whether the exclusive pre-write mean and variance predict the current innovation. In the Gaussian model, this integral is available exactly and yields the scalar coordinate terms summed in Equation (181); no latent sampling or variational bound is needed to evaluate it.

Bayes' rule gives the exact variational identity

$$-\text{ELBO}_{t,j}(q_{t,j}) = \mathcal{L}_{\text{pred},t,j} + \text{KL}(q_{t,j} \parallel p(\mathbf{s}_{t,j}^* | v_{t,j}, \mathcal{F}_{t-1}, \mathbf{k}_t)). \quad (189)$$

Thus the negative ELBO upper-bounds the predictive marginal NLL and equals it when $q_{t,j}$ is the exact posterior. A diagonal or isotropic variational family generally leaves a nonnegative gap

equal to its reverse-KL error. Importantly, it is the complete negative ELBO, not the expected reconstruction term in Equation (187), that equals the predictive NLL at the exact posterior.

The analogy to a VAE is therefore limited. The latent memory column plays the role of the VAE latent variable, the value coordinate plays the role of the observation, and $q_{t,j}$ plays an encoder-like role. Here, however, inference is an online filtering update with a history-dependent prior, the linear-Gaussian predictive marginal is analytic, and both $\mathbf{k}_t$ and $\mathbf{v}_t$ are learned internal representations rather than observed language tokens. Consequently, the predictive NLL is an internal probabilistic-consistency and covariance-calibration objective, not a variational lower bound on the language-model likelihood.

Finally, a single Gaussian likelihood assigns the same multiplier to the mean-residual and posterior-variance terms in Equation (187). The unequal ($s_{\text{write}}, s_{\text{protect}}$) objective above remains a valid decision-oriented criterion, but it is not the ELBO of one coherent Gaussian observation model. If r_t is learned, the $\log r_t$ normalizer must be retained; otherwise increasing r_t can reduce both quadratic terms without paying the corresponding likelihood cost.

10.4 A VAE-style auxiliary loss on the latent memory

The preceding identity also suggests an explicit variational auxiliary loss on the complete memory state. Under the columnwise model, define the post-write approximation and pre-write prior over the latent memory matrix by

$$q_t(\mathbf{S}^*) = \prod_{j=1}^{d_v} \mathcal{N}(\mathbf{s}_{:,j}^*; \mathbf{s}_{t,j}, \mathbf{P}_t), \quad \hat{\pi}_t(\mathbf{S}^*) = \prod_{j=1}^{d_v} \mathcal{N}(\mathbf{s}_{:,j}^*; \hat{\mathbf{s}}_{t,j}, \hat{\mathbf{P}}_t). \quad (190)$$

Here $\mathbf{S}_t$ is the post-write posterior mean represented by the recurrent memory, whereas $\hat{\mathbf{S}}_t$ is its exclusive pre-write prediction. With

$$p(\mathbf{v}_t \mid \mathbf{S}^*, \mathbf{k}_t) = \mathcal{N}(\mathbf{v}_t; (\mathbf{S}^*)^\top \mathbf{k}_t, r_t \mathbf{I}_{d_v}), \quad (191)$$

the VAE-style memory loss is the negative one-step ELBO

$$\mathcal{J}_{\text{mem},t} = \underbrace{\mathbb{E}_{q_t}[-\log p(\mathbf{v}_t \mid \mathbf{S}^*, \mathbf{k}_t)]}_{\text{memory reconstruction}} + \underbrace{\text{KL}(q_t \parallel \hat{\pi}_t)}_{\text{regularization to the pre-write memory}}. \quad (192)$$

No stochastic memory sample is required. Because both distributions and the observation model are Gaussian, the two terms are available in closed form. The reconstruction term is

$$\mathbb{E}_{q_t}[-\log p(\mathbf{v}_t \mid \mathbf{S}^*, \mathbf{k}_t)] = \frac{1}{2r_t} \left[\left\| \mathbf{v}_t - \mathbf{S}_t^\top \mathbf{k}_t \right\|_2^2 + d_v \mathbf{k}_t^\top \mathbf{P}_t \mathbf{k}_t \right] + \frac{d_v}{2} \log(2\pi r_t), \quad (193)$$

and the KL term is

$$\begin{aligned} \text{KL}(q_t \parallel \hat{\pi}_t) = \frac{1}{2} \Bigg\{ & \text{tr} \left[\hat{\mathbf{P}}_t^{-1} (\mathbf{S}_t - \hat{\mathbf{S}}_t) (\mathbf{S}_t - \hat{\mathbf{S}}_t)^\top \right] \\ & + d_v \left[\text{tr}(\hat{\mathbf{P}}_t^{-1} \mathbf{P}_t) - d_k + \log \frac{\det \hat{\mathbf{P}}_t}{\det \mathbf{P}_t} \right] \Bigg\}. \end{aligned} \quad (194)$$

For diagonal or isotropic covariances, the traces and log determinants reduce to elementwise sums, so this loss has no sampling estimator and does not require materializing a dense covariance. The

mean-dependent terms also simplify under the rank-one KDN write. With $\delta_t = v_t - \hat{S}_t^\top k_t$ and $S_t - \hat{S}_t = \kappa_t \delta_t^\top$,

$$\begin{aligned} \|v_t - S_t^\top k_t\|_2^2 &= \left(1 - k_t^\top \kappa_t\right)^2 \|\delta_t\|_2^2, \\ \text{tr}\left[\hat{P}_t^{-1}(S_t - \hat{S}_t)(S_t - \hat{S}_t)^\top\right] &= \|\delta_t\|_2^2 \kappa_t^\top \hat{P}_t^{-1} \kappa_t. \end{aligned} \quad (195)$$

Diagonal and Isotropic KDN can therefore evaluate the auxiliary using their existing states and token-local reductions.

The two terms have the same roles as in a VAE. The reconstruction term makes the post-write memory explain the current value and penalizes posterior uncertainty along the observed key. The KL term limits how far the posterior mean and covariance can move from their pre-write predictions. Because the posterior is allowed to depend on the current value, the reconstruction term alone can reward copying that value into memory; the KL term is what turns the complete objective into a valid variational bound.

Summing Equation (189) over value coordinates gives

$$\mathcal{J}_{\text{mem},t} = \mathcal{L}_{\text{pred},t} + \sum_{j=1}^{d_v} \text{KL}(q_{t,j} \| p(s_{t,j}^* | v_{t,j}, \mathcal{F}_{t-1}, k_t)). \quad (196)$$

Thus this is not an independent copy of the predictive calibration loss. It reduces to the predictive NLL when the retained posterior is exact; for Diagonal or Isotropic KDN it additionally penalizes the full posterior-approximation gap. Under the ordinary ELBO projection, this gap is the reverse-KL error imposed by the restricted covariance family. Under a split $(s_{\text{write}}, s_{\text{protect}})$ update with $s_{\text{protect}} \neq s_{\text{write}}$, it also penalizes the deliberate deviation from the ordinary Gaussian posterior and may therefore weaken the intended protection against future overwrites. The predictive NLL is the more direct choice when the goal is only to calibrate the pre-write innovation variance, whereas $\mathcal{J}_{\text{mem},t}$ is appropriate when the goal also includes regularizing the post-write memory approximation toward the coherent Gaussian posterior.

A dimension-normalized auxiliary objective can be added to language-model training as

$$\mathcal{L} = \mathcal{L}_{\text{LM}} + \lambda_{\text{mem}} \mathbb{E}_{t,h} \left[\frac{1}{d_v} \mathcal{J}_{\text{mem},t,h} \right]. \quad (197)$$

This term should initially be tested with a small λ_{mem} , a fixed covariance gauge, and all Gaussian normalizers retained. Full gradients can also change the scale and geometry of the learned keys and values rather than only calibrating uncertainty, so they should be compared with a route that treats the innovation and key geometry as fixed targets. Because $\mathcal{J}_{\text{mem},t}$ already contains $\mathcal{L}_{\text{pred},t}$, the two auxiliary losses should normally be evaluated separately rather than summed without explicit reweighting.

10.5 Calibration diagnostics and optimization caveat

Under a correctly specified predictive model, the normalized innovation squared

$$\text{NIS}_t = \frac{\|\delta_t\|_2^2}{\Delta_t} \quad (198)$$

has a $\chi_{d_v}^2$ distribution, with mean d_v and variance $2d_v$. Accordingly, useful diagnostics are NIS_t/d_v , the predictive NLL, and calibration bins comparing predicted Δ_t with the empirical quantity

$\|\delta_t\|_2^2/d_v$. These should be stratified by layer, head, position, and key similarity. A mean normalized innovation near one is necessary but not sufficient: averages can hide systematic overconfidence on repeated keys and underconfidence on novel keys.

The predictive NLL also breaks the covariance-only scale symmetry when the innovation units are held fixed, because rescaling $(\hat{\mathbf{P}}_t, r_t)$ changes Δ_t and hence changes the score. In the neural mixer, however, both $\mathbf{v}_t$ and $\mathbf{k}_t$ are learned internal representations rather than externally observed quantities. They can change scale or geometry together with the uncertainty heads and downstream projections. The resulting objective is therefore best described as an internal probabilistic consistency loss, not automatically as a calibrated likelihood of the language data.

This distinction suggests a staged experiment. First evaluate the NLL and normalized innovation without adding them to training. If they are optimized, compare: (i) full gradients through the internal representation; (ii) a calibration-only route in which the realized innovation and key geometry are treated as fixed targets and gradients are sent only to the uncertainty parameters; and (iii) the language-modeling objective alone. Fixing a covariance gauge, preferably $r = 1$ and $\mu = 1$, remains useful in all three cases. An auxiliary loss

$$\mathcal{L} = \mathcal{L}_{\text{LM}} + \lambda_{\text{pred}} \mathbb{E}_{t,h} [\overline{\mathcal{L}}_{\text{pred},t,h}] \quad (199)$$

should begin with small λ_{pred} and be judged not only by predictive NLL but also by language-modeling loss, write strength, covariance quantiles, and downstream recall. A lower internal NLL is evidence that the noise heads predict their own innovation statistics more consistently; it is not by itself evidence of a better language model or a more faithful full covariance approximation.

11 Causal Associative Replay as a Direct Overwrite Objective

The predictive NLL evaluates whether pre-write uncertainty matches the current innovation, and the memory ELBO additionally evaluates the quality of the post-write posterior approximation. Neither objective directly states that an association written in the past should remain recoverable after intervening writes. A causal replay objective can provide this missing behavioral signal: at time t , sample an earlier token $\tau < t$, query the current memory with its key $\mathbf{k}_\tau$, and compare the result with a detached target from time τ .

11.1 Raw replay mixes writing and retention

The most direct construction uses the old value itself,

$$\tilde{\mathbf{v}}_{\tau|t} = \mathbf{S}_t^\top \mathbf{k}_\tau, \quad \mathcal{L}_{\text{raw-replay}} = \mathbb{E}_{t,\tau \sim \mathcal{Q}_t} \left[\frac{1}{d_v} \|\tilde{\mathbf{v}}_{\tau|t} - \text{sg}(\mathbf{v}_\tau)\|_2^2 \right], \quad (200)$$

where $\mathcal{Q}_t$ is a causal distribution over past indices and $\text{sg}(\cdot)$ denotes stop-gradient. This objective is causal and adds no inference cost, but it conflates three effects: whether the association was written accurately at τ , whether the transition intentionally decayed it, and whether later writes interfered with it.

To see the first issue, define the post-write read and effective gain at the original token by

$$\mathbf{y}_\tau^+ = \mathbf{S}_\tau^\top \mathbf{k}_\tau, \quad g_\tau = \mathbf{k}_\tau^\top \boldsymbol{\kappa}_\tau. \quad (201)$$

The KDN update gives

$$\mathbf{y}_\tau^+ = \hat{\mathbf{S}}_\tau^\top \mathbf{k}_\tau + g_\tau \boldsymbol{\delta}_\tau, \quad \mathbf{v}_\tau - \mathbf{y}_\tau^+ = (1 - g_\tau) \boldsymbol{\delta}_\tau. \quad (202)$$

Consequently, raw replay pressure can increase g_τ toward one in order to make the initial write reproduce $\mathbf{v}_\tau$ more exactly. This may strengthen the very current-token overwrite that the experiment is intended to diagnose. It also asks a finite-capacity memory to preserve every sampled internal value, including associations that the language-modeling objective may prefer to forget.

11.2 Post-write retention consistency

A cleaner overwrite target preserves what was actually stored rather than requiring perfect initial reconstruction. Define

$$\mathcal{L}_{\text{retain}} = \mathbb{E}_{t,\tau \sim Q_t} \left[w_{t,\tau} \frac{\|\mathbf{S}_t^\top \text{sg}(\mathbf{k}_\tau) - \text{sg}(\mathbf{y}_\tau^+)\|_2^2}{\|\text{sg}(\mathbf{y}_\tau^+)\|_2^2 + \epsilon} \right]. \quad (203)$$

This loss does not change the target when the model later changes its key or value representation, and the normalization prevents the scale of the internal value space from dominating the loss. The weight $w_{t,\tau}$ may prioritize associations that were actually written, for example using a detached function of g_τ or of the original write magnitude. Compared with Equation (200), this construction separates initial write quality from subsequent retention. It can still oppose intentional transition decay, because its target is the undiminished post-write read.

If the goal is to isolate interference from later writes while preserving the model’s chosen decay, define the transition product

$$\Phi_{t \leftarrow \tau} = \mathbf{D}_t \mathbf{D}_{t-1} \cdots \mathbf{D}_{\tau+1} \quad (204)$$

and the decay-only counterfactual

$$\mathbf{y}_{\tau \rightarrow t}^{\text{decay}} = (\Phi_{t \leftarrow \tau} \mathbf{S}_\tau)^\top \mathbf{k}_\tau. \quad (205)$$

Replacing $\mathbf{y}_\tau^+$ in Equation (203) with a stopped-gradient copy of $\mathbf{y}_{\tau \rightarrow t}^{\text{decay}}$ penalizes only the deviation caused by intervening writes. This is the most specific test of overwrite, but it requires retaining or recomputing the anchor state and its decay-only trajectory. The simpler post-write target requires storing only $(\mathbf{k}_\tau, \mathbf{y}_\tau^+)$ and is therefore a cheaper first experiment.

11.3 A local interference surrogate

An even cheaper objective measures the disturbance produced by the current write at a sampled past address. Since

$$\mathbf{S}_t - \widehat{\mathbf{S}}_t = \boldsymbol{\kappa}_t \boldsymbol{\delta}_t^\top, \quad (206)$$

the instantaneous change in the read at $\mathbf{k}_\tau$ is exactly

$$(\mathbf{S}_t - \widehat{\mathbf{S}}_t)^\top \mathbf{k}_\tau = \boldsymbol{\delta}_t^\top (\boldsymbol{\kappa}_t^\top \mathbf{k}_\tau). \quad (207)$$

This yields the sampled penalty

$$\mathcal{L}_{\text{interfere}} = \mathbb{E}_{t,\tau \sim Q_t} \left[w_{t,\tau} \frac{(\mathbf{k}_\tau^\top \boldsymbol{\kappa}_t)^2 \|\boldsymbol{\delta}_t\|_2^2}{\|\text{sg}(\mathbf{y}_\tau^+)\|_2^2 + \epsilon} \right]. \quad (208)$$

Only the past key and a normalization statistic need to be cached. This surrogate directly penalizes overlap between the current write direction and past addresses, but it treats every change as harmful. Repeated keys can legitimately revise an old association, so the sampling or weighting rule must identify associations intended to persist.

11.4 Sampling, optimization, and relation to calibration

Evaluating all pairs (τ, t) would be quadratic. A practical implementation can retain a small number of detached anchors per chunk and evaluate them at later chunk boundaries, turning the additional reads into a small batched matrix multiplication. A log-uniform lag distribution covers both local and long-range retention. Mixing these samples with hard candidates selected from a small pool by key similarity concentrates supervision on pairs most likely to interfere without forming the full pairwise similarity matrix.

The recommended first experiment is the post-write consistency loss in Equation (203), with one or a few anchors per chunk, detached keys and targets, a small loss weight, and explicit reporting by recall distance. The raw value loss should be kept as an ablation because it combines stronger initial writing with retention. The local interference loss is useful as a low-cost diagnostic or regularizer, but it does not measure accumulated corruption from several individually small writes. To isolate changes after τ , the auxiliary gradient should also be truncated at the anchor state; otherwise the loss is an end-to-end retention objective that may change how the original association is stored. Likewise, an uncertainty-only version of Equation (208) should detach δ_t and both keys, routing gradients only through κ_t , so that the model cannot reduce the penalty merely by shrinking values or rotating the key representation.

With a fixed information scale of one, replay supplies a decision-oriented retention signal that the coherent Gaussian ELBO does not contain. It does not calibrate $\mathbf{P}_t$; instead, it states which past associations should remain useful. The two goals can be combined as

$$\mathcal{L} = \mathcal{L}_{\text{LM}} + \lambda_{\text{pred}}\mathcal{L}_{\text{pred}} + \lambda_{\text{retain}}\mathcal{L}_{\text{retain}}, \quad (209)$$

but the losses should first be ablated separately. Predictive NLL rewards uncertainty that matches one-step errors, whereas replay rewards memory that survives interference. If a diagonal model cannot improve both simultaneously, that conflict is evidence that one covariance state is being asked to serve incompatible calibration and protection roles. In that case, separating a calibrated covariance from a protection state is more principled than forcing either loss to absorb the mismatch. A clean factorial experiment should compare unit and d_k -scaled protection with replay disabled and enabled, reporting predictive NLL or NIS, replay error by lag, effective-write strength, language-modeling loss, and downstream recall.

12 Conclusion

The diagonal KDN has a genuine common-scale non-identifiability in (ω, r, μ) , but the completed noisy-MQAR experiments show that random initial anisotropy is the dominant observed instability. The appropriate design is therefore twofold: start the gain isotropically, and fix a covariance unit—preferably $r = 1$ and $\mu = 1$ —so the model learns only uncertainty ratios that affect its predictions.

Process noise should be constrained by that gauge, not removed by it. Section 7 shows that ω is not merely the third carrier of the scale symmetry: it sets the stationary plasticity floor $\Theta(\sqrt{\omega r})$, and without it the layer degenerates into recursive least squares with a gain that decays as $d_k/(st)$. The fidelity of the diagonal approximation, meanwhile, is governed by key geometry through Equation (112) and is unaffected by ω either way. Fixing r and μ while keeping a positive, isotropically initialized ω is therefore the design that removes the flat direction without removing the mechanism.

Finally, reverse-KL overconfidence and anisotropic overwrite require separate controls. The proposed $B = 2$ channel cap is a scan-compatible guardrail against unbounded current-write

geometry; it should be evaluated alongside off-key shrinkage and explicit effective-write caps rather than interpreted as a more faithful covariance approximation.